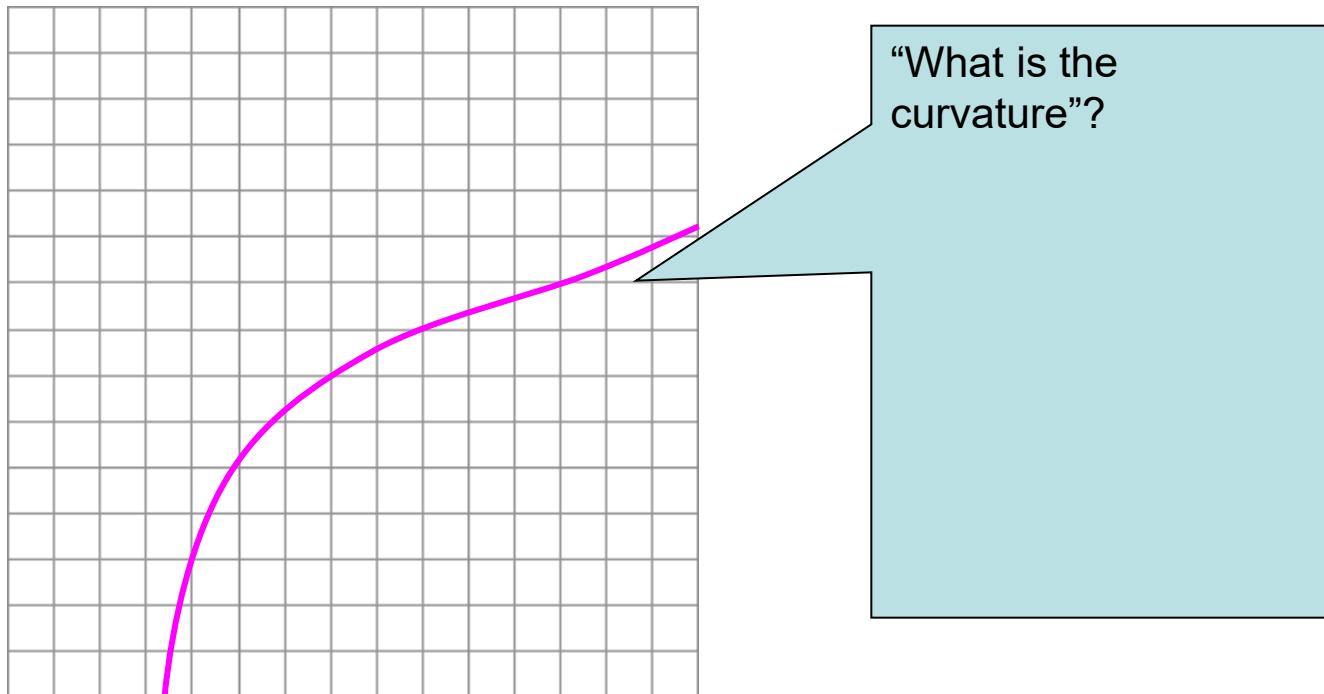


# Algebras and Calculus of Numbers and Shapes

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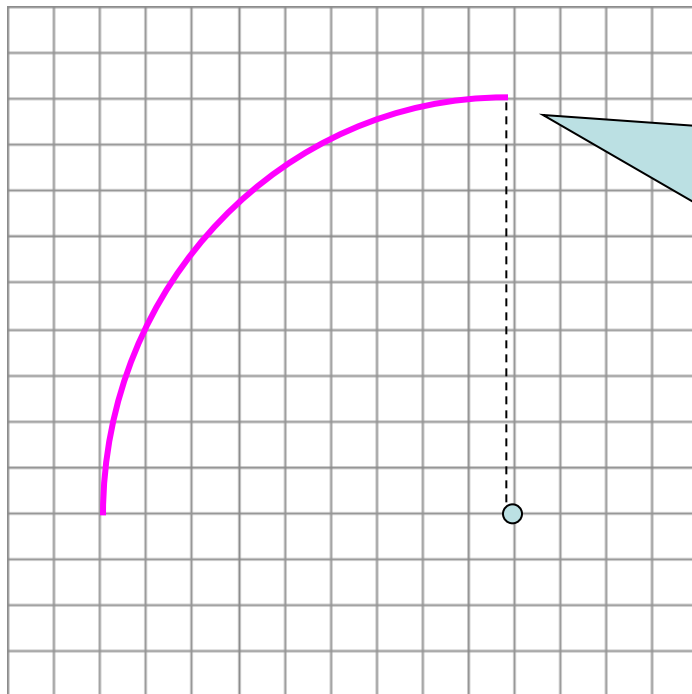


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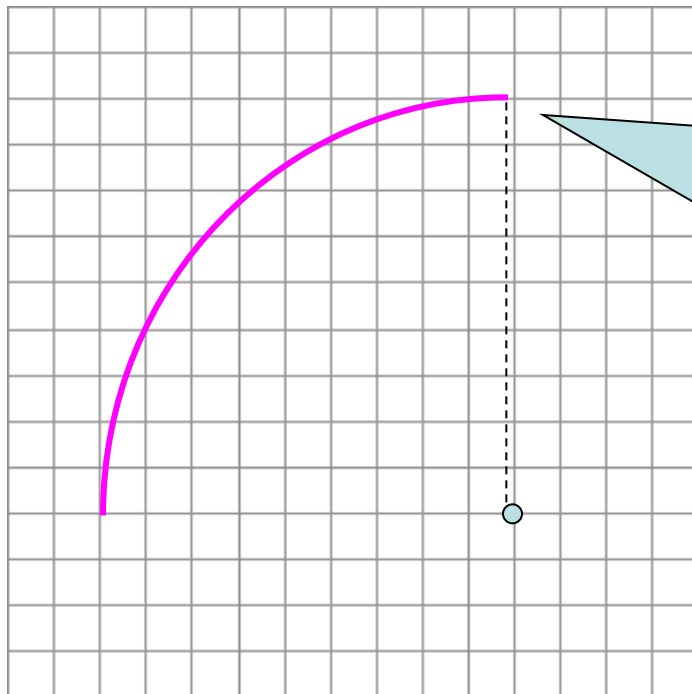
$$\text{Curvature} = 1/\text{radius}$$

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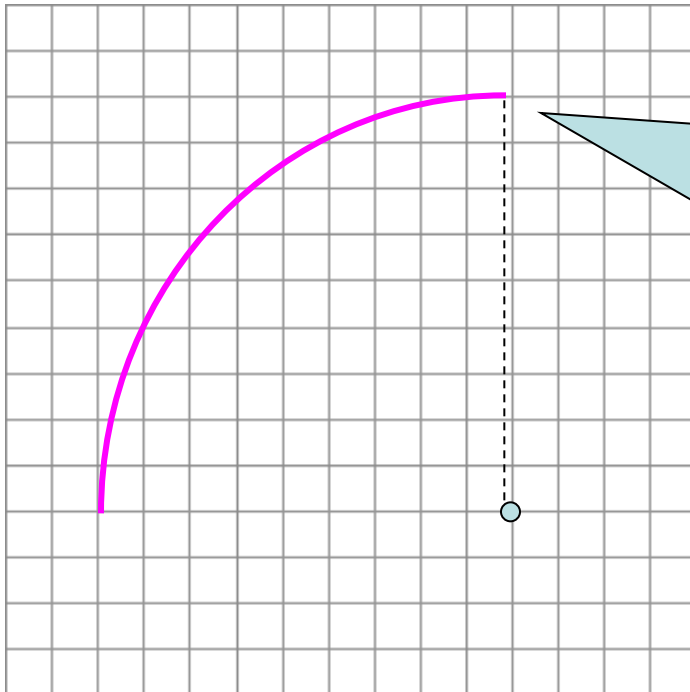
- Position varies
- Direction varies
- Direction to the center varies

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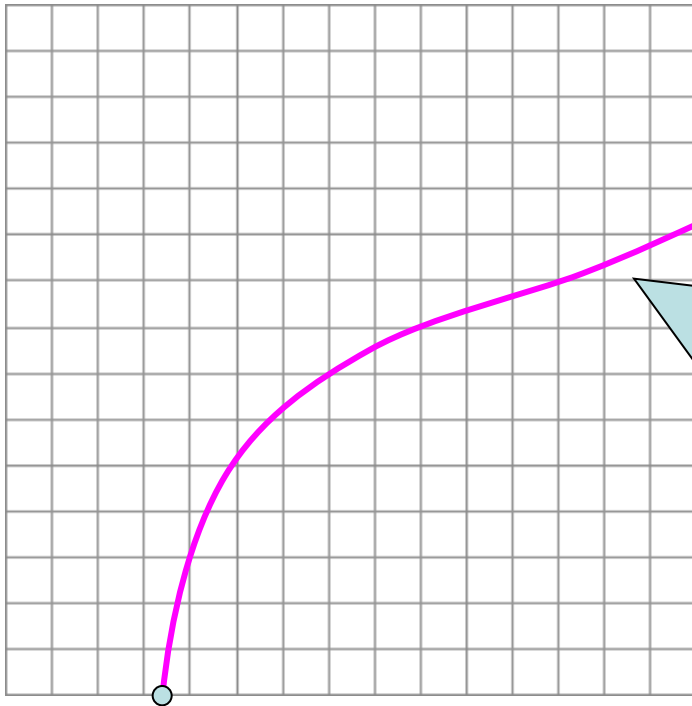
The fact that the curvature is static indicates something “degenerate” about the shape

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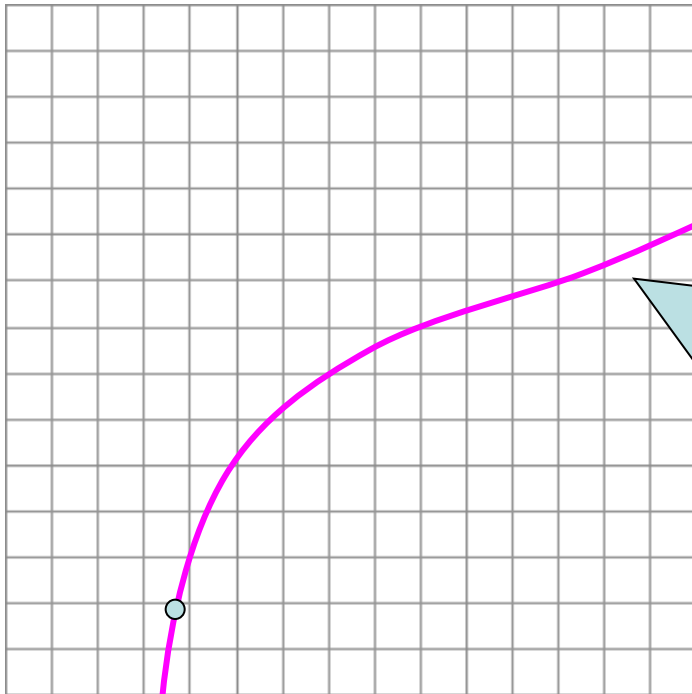
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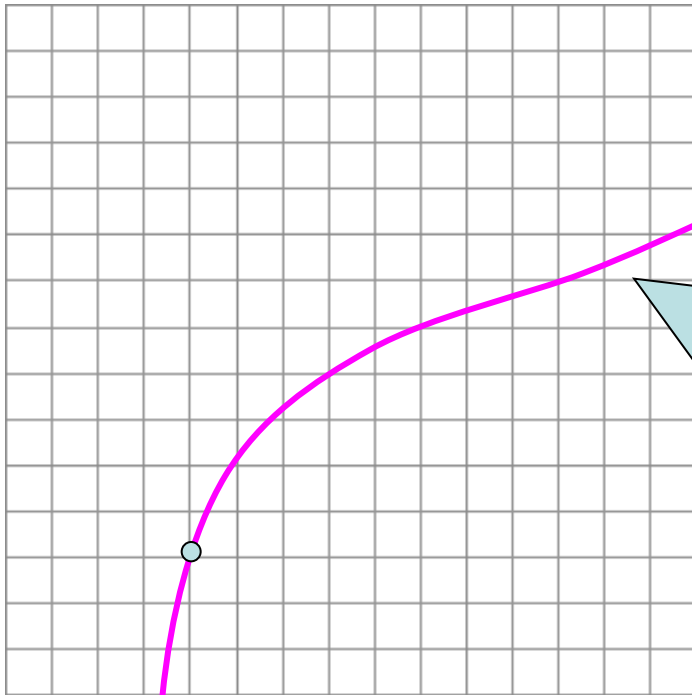
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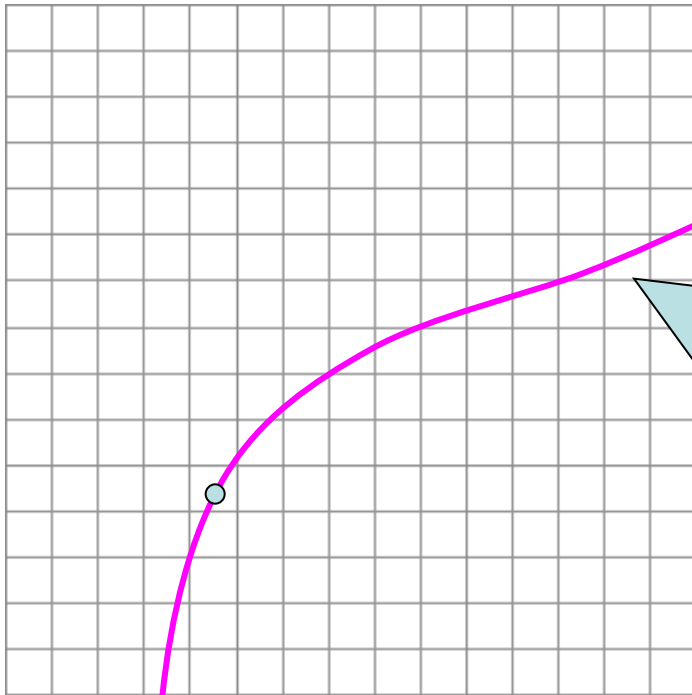
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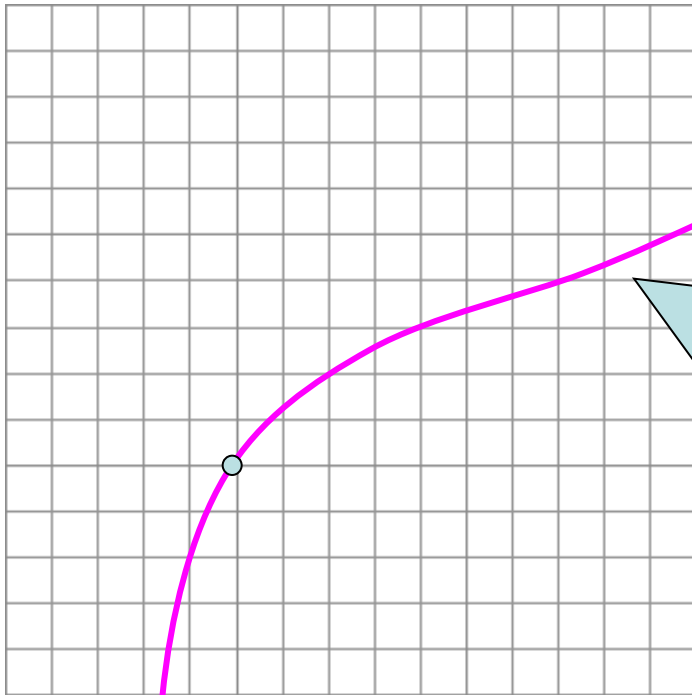


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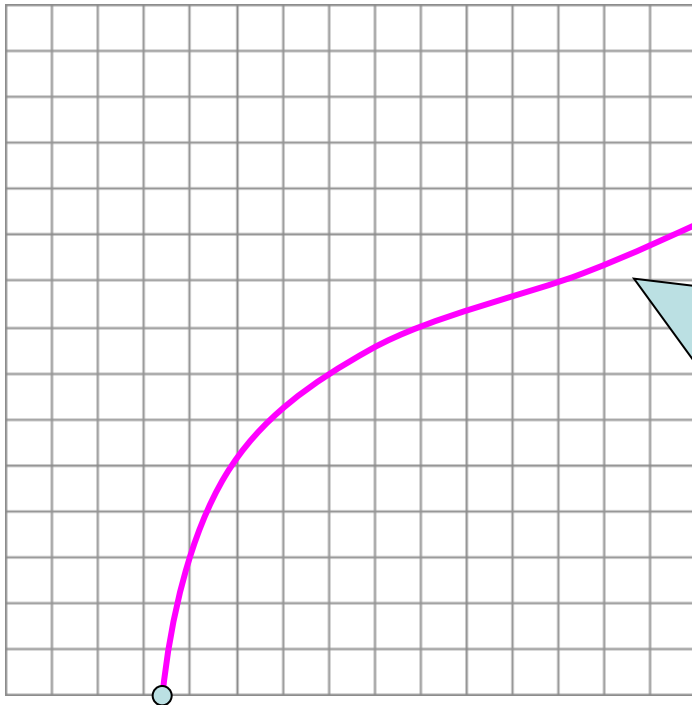
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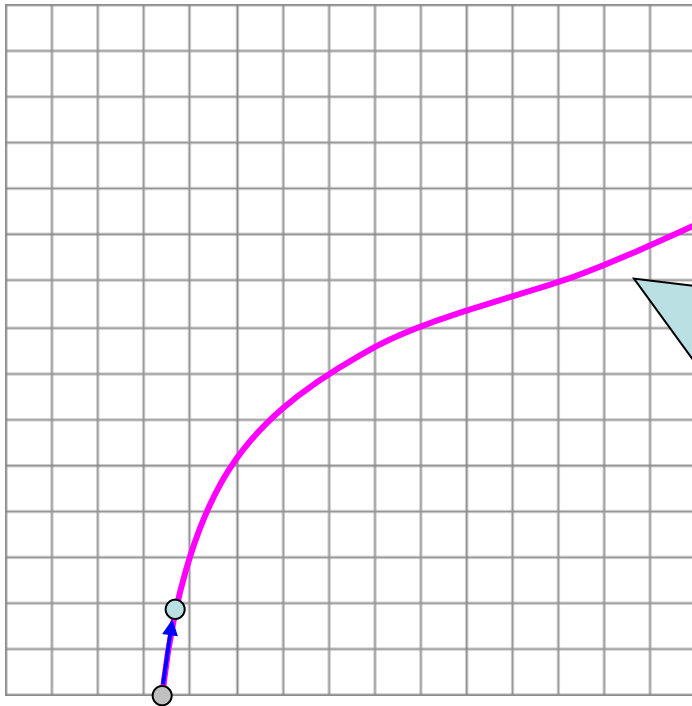
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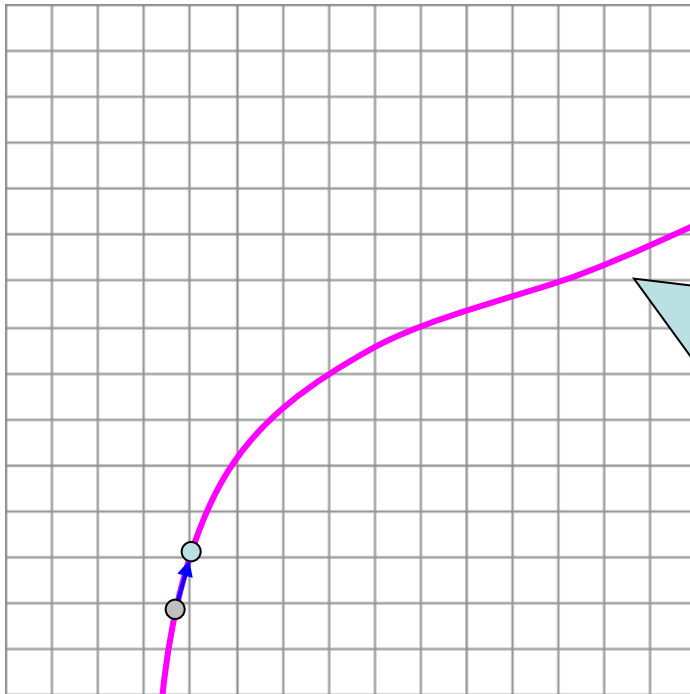
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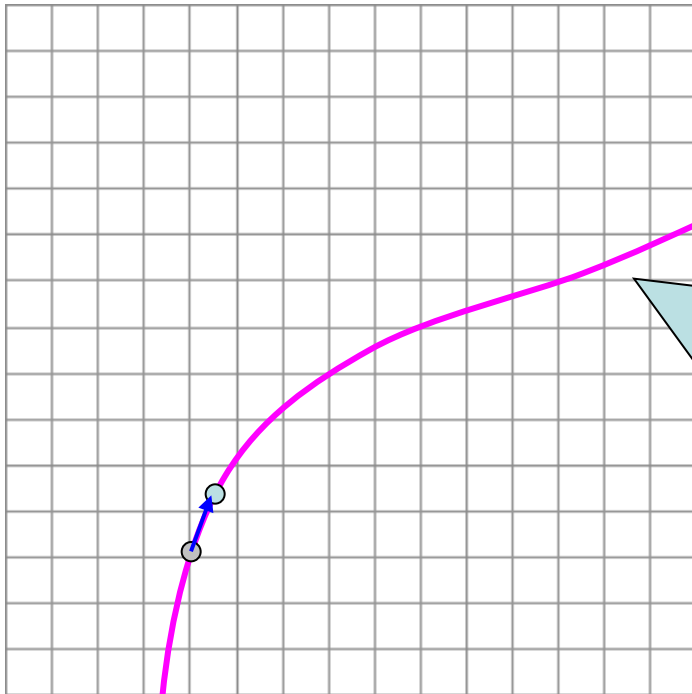
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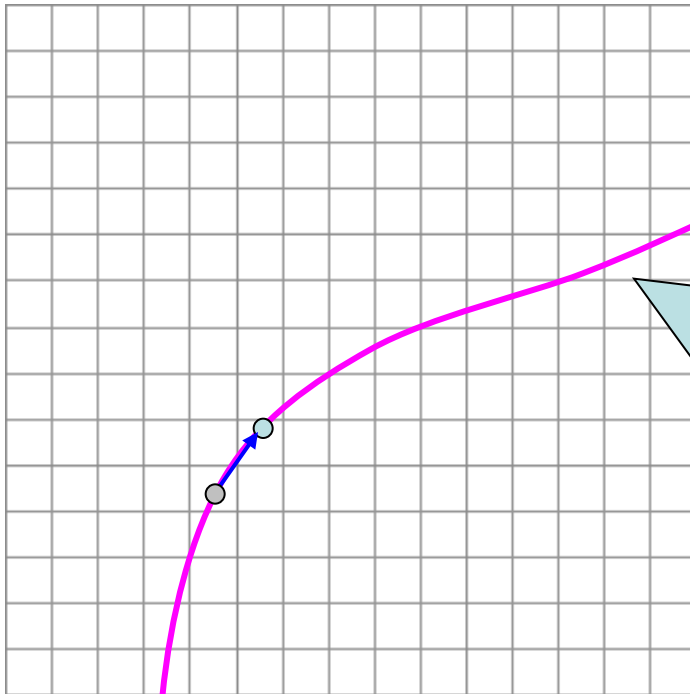
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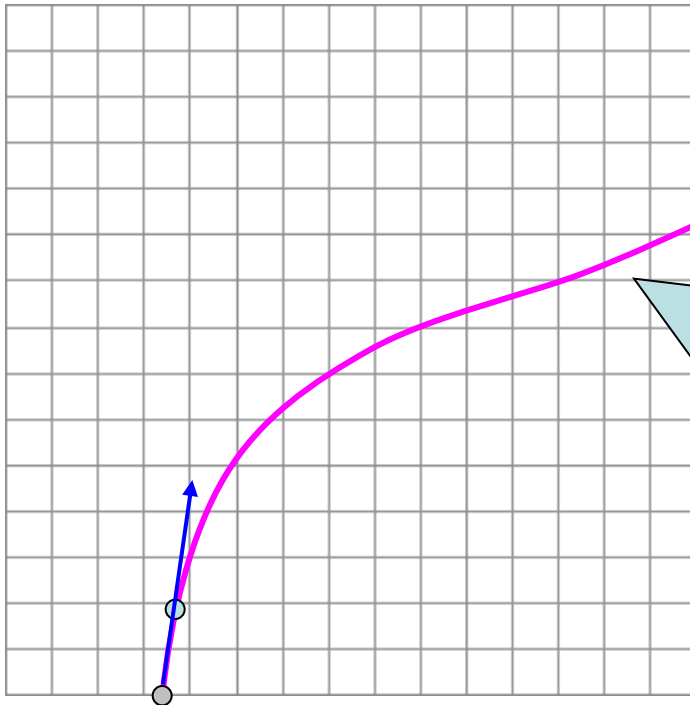
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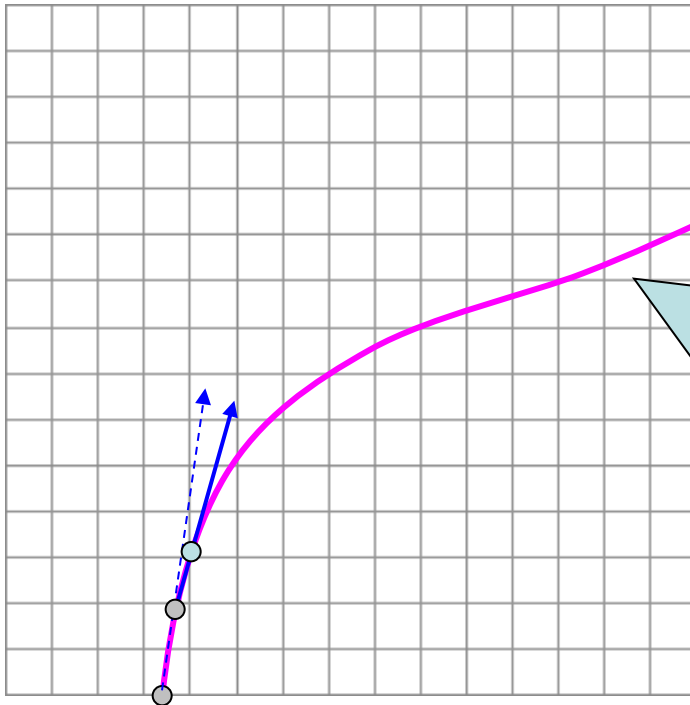
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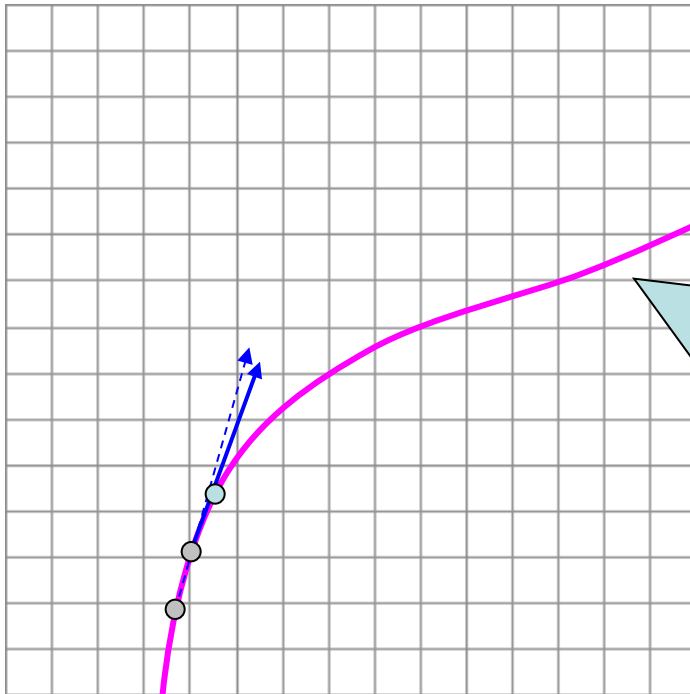


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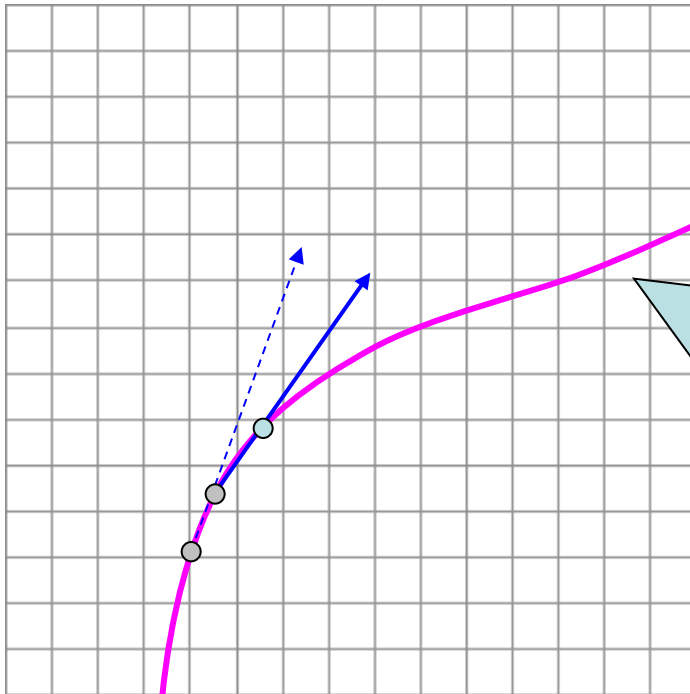
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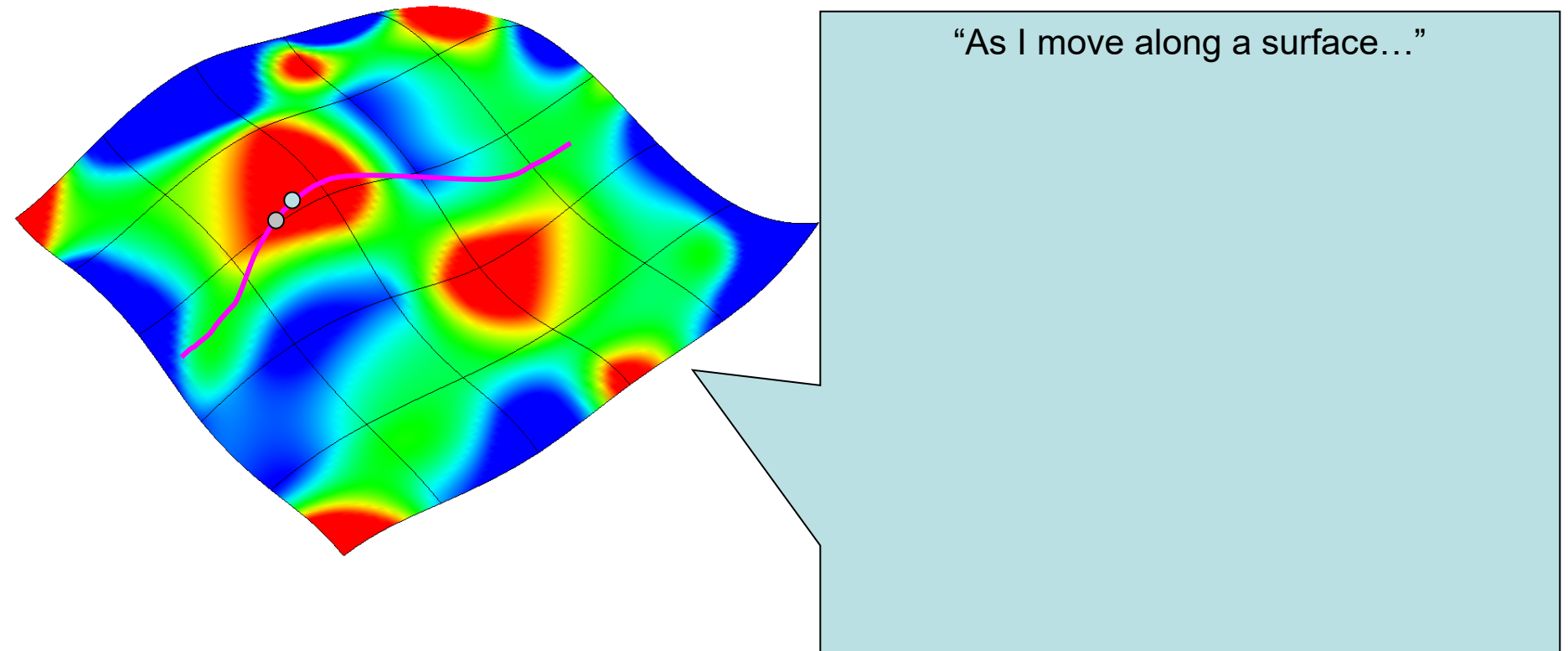
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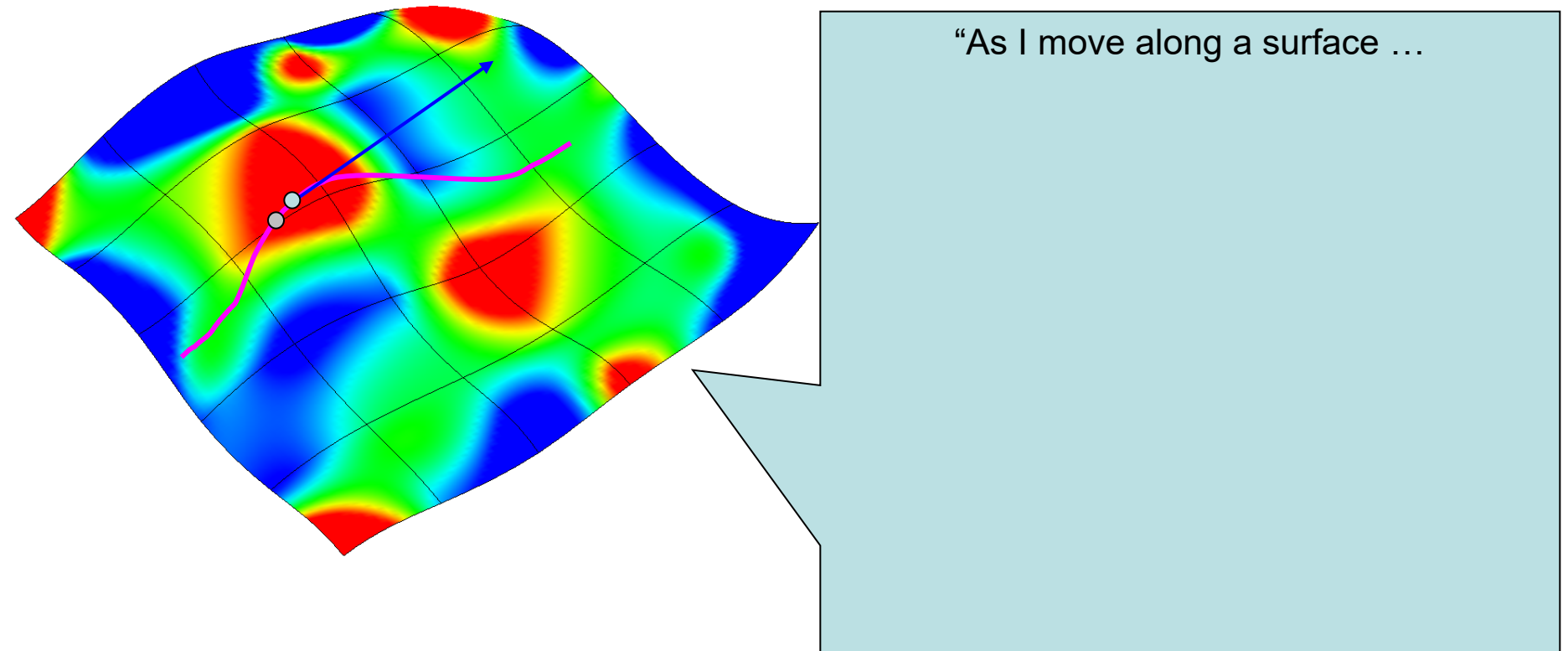
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Example: surface curvature



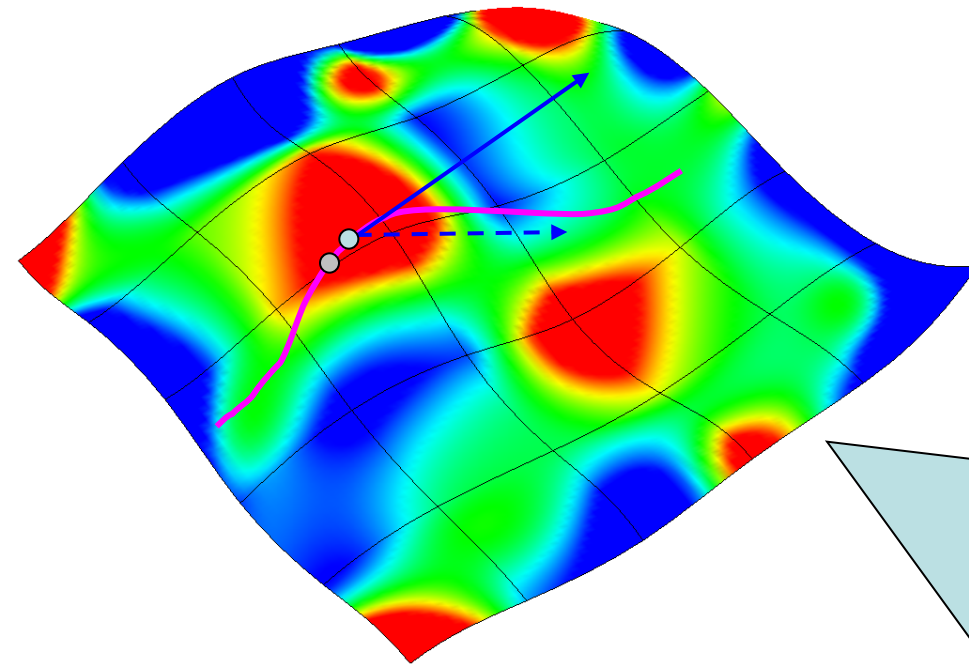
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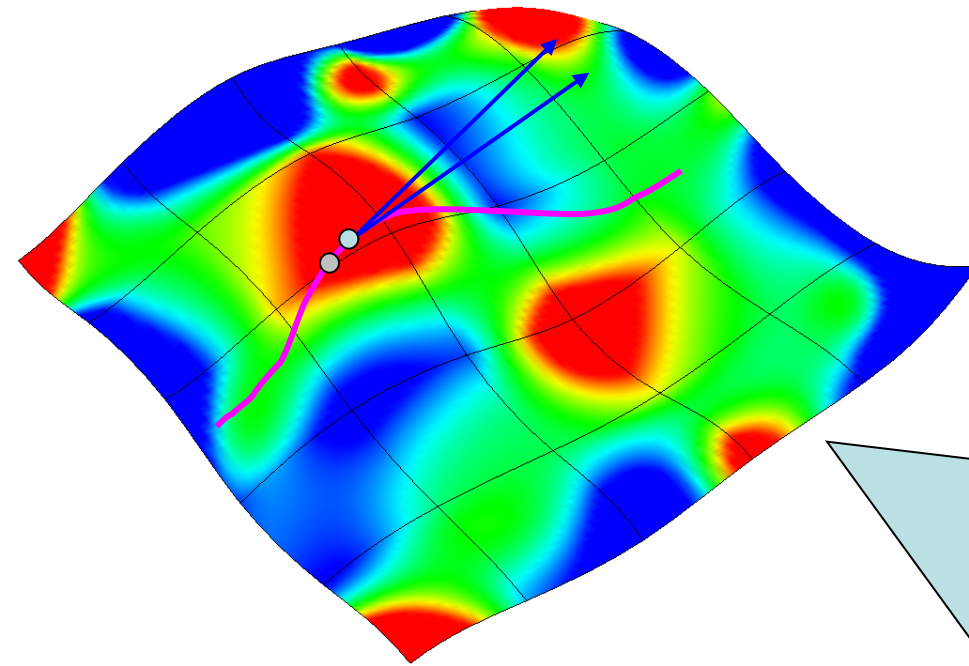


“As I move along a surface ...

My change in position varies **with** the  
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# Algebras and Calculus of Numbers and Shapes

Example: surface curvature



“As I move along a surface ...

My change in position varies **with** the  
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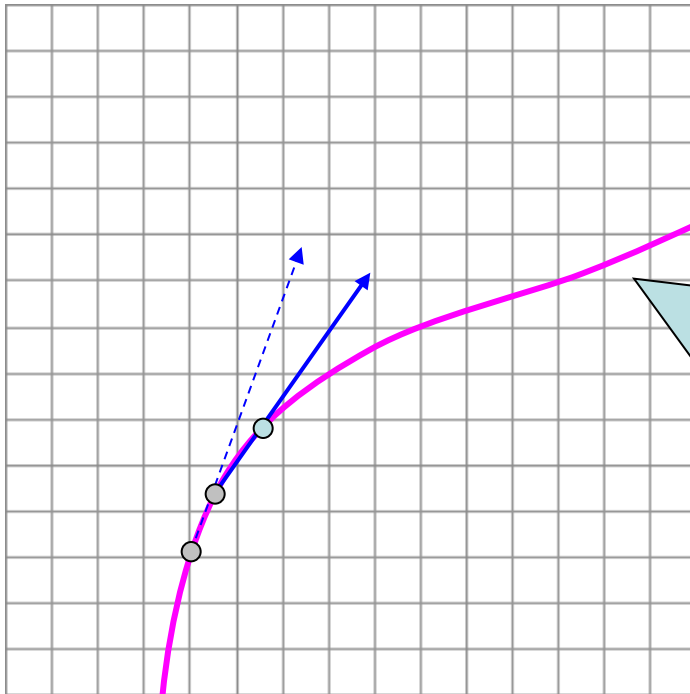
and **within** the surface”

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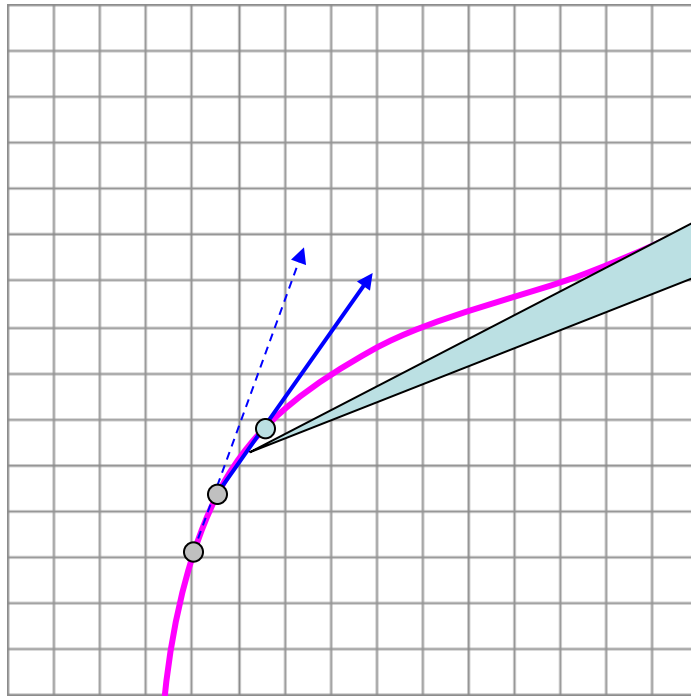
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# Calculus

See the world from 2 directions:  
Myopically

Differentiation



*“What is the rate of change of distance  
as I move along the curve?”*

*(and many other characteristics)*



# Calculus

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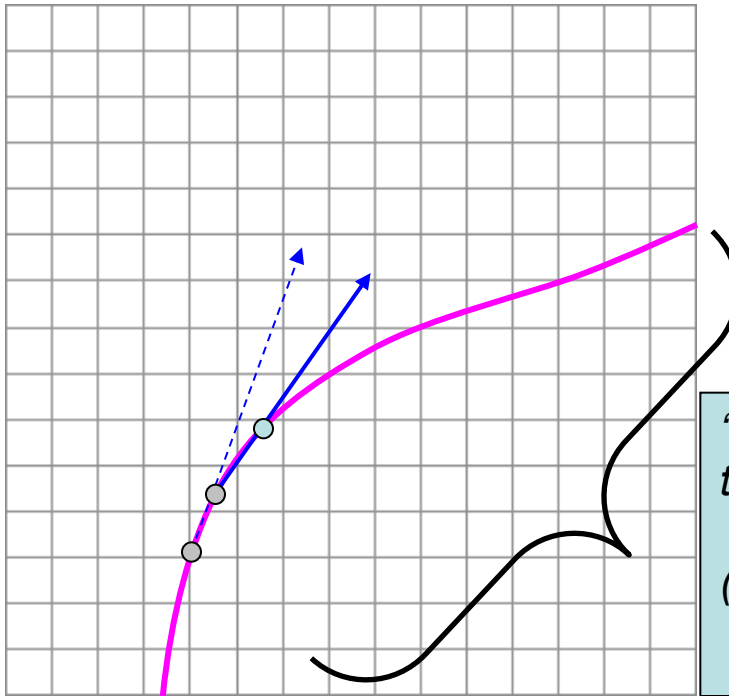
Globally

Differentiation

Integration

*“What is the rate of change of distance  
as I move along the curve?”*

*(and many other characteristics)*

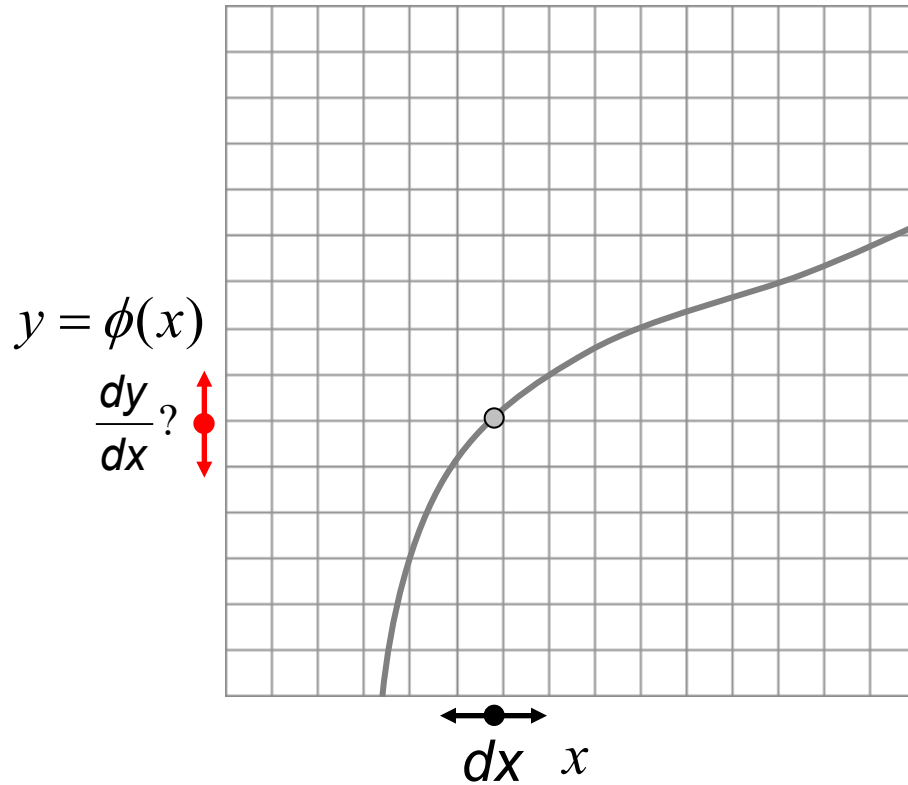


*“What is the overall distance (length) of  
the curve?”*

*(and many other characteristics)*

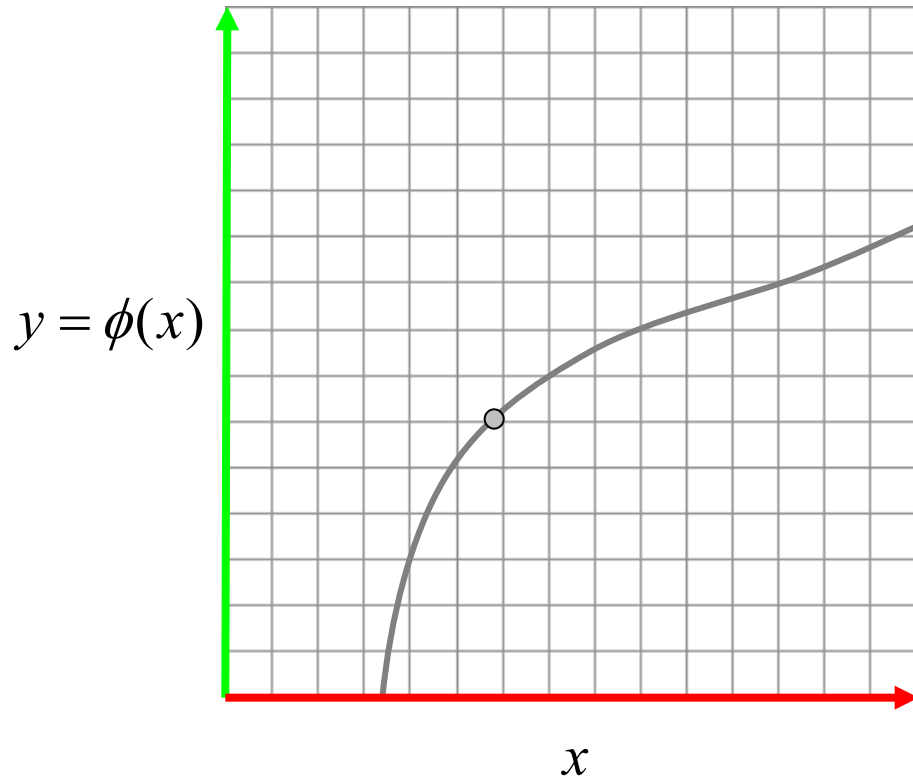
# Parametric Curves

*old  
explicit function*

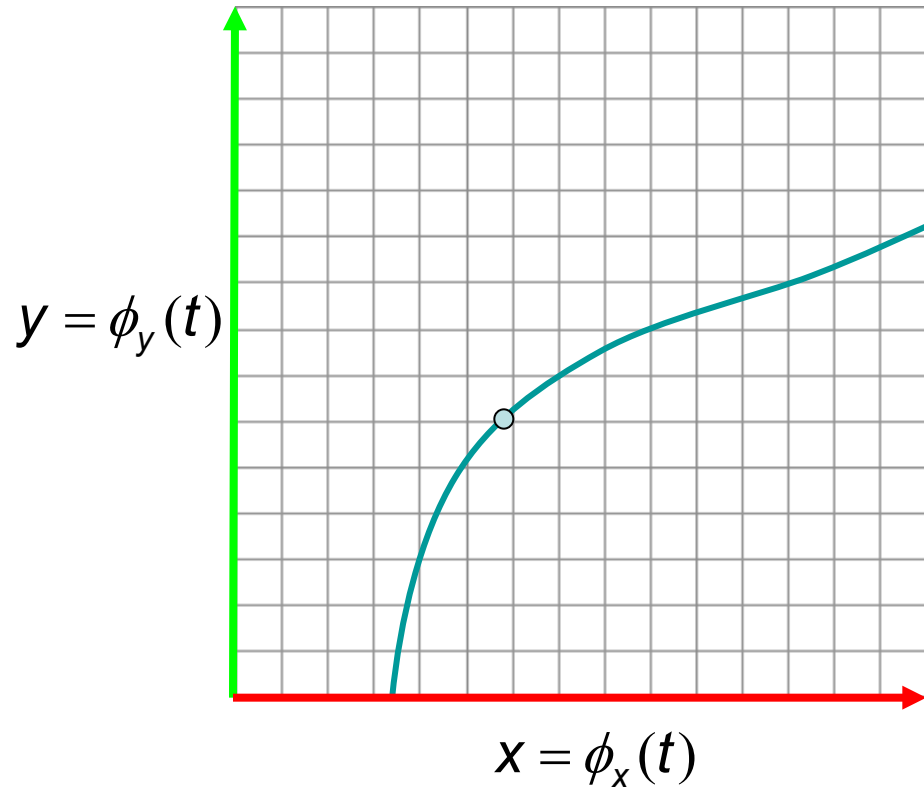
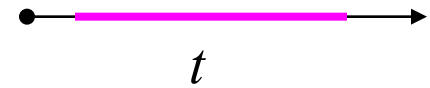


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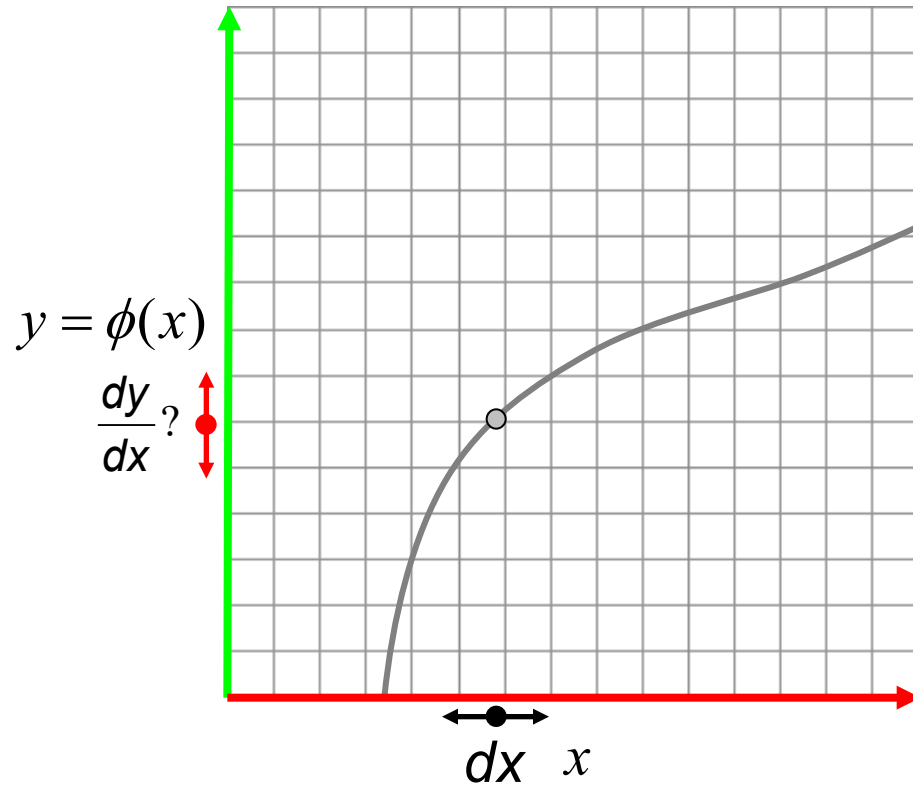


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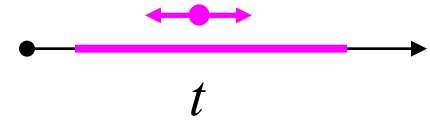


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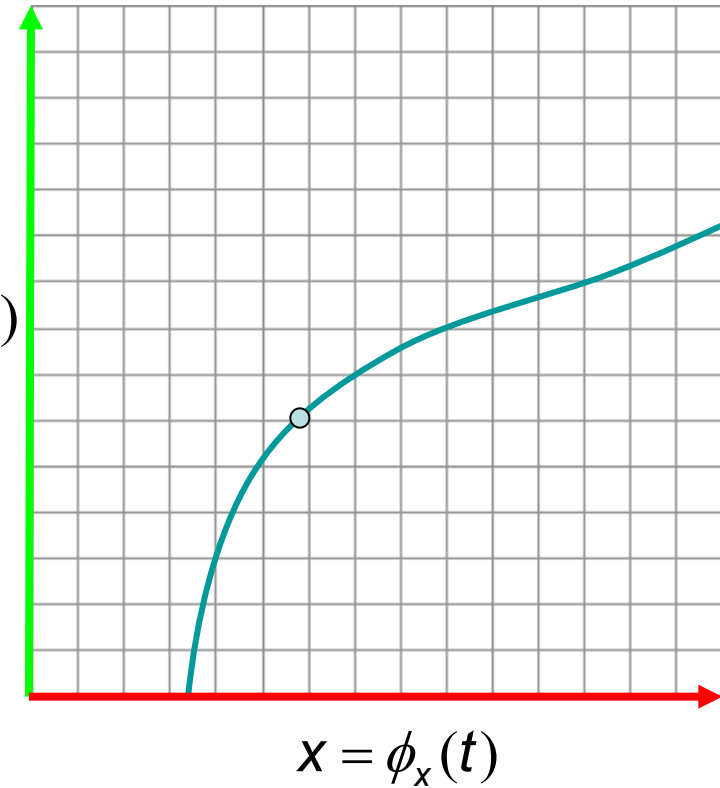
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 $dt$*

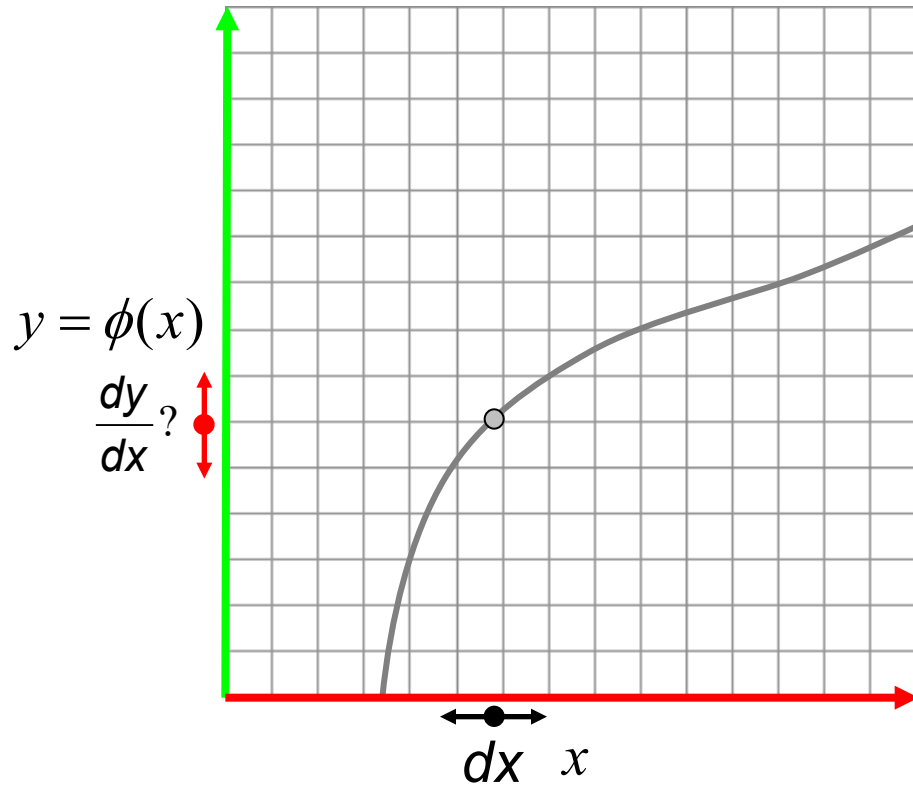


$$y = \phi_y(t)$$

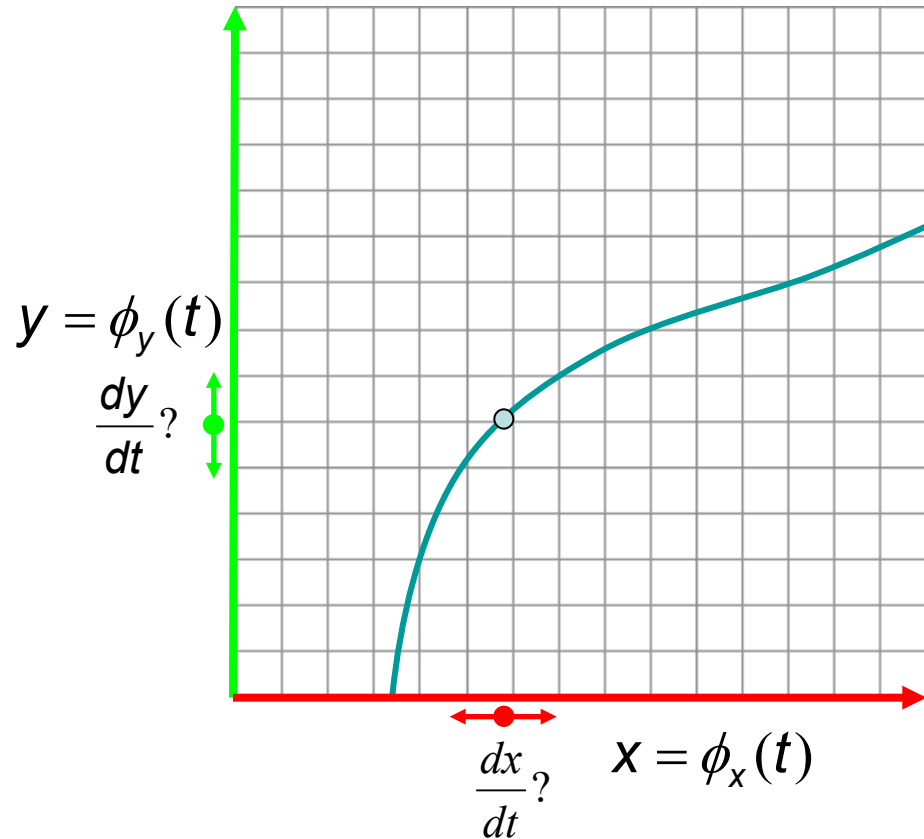
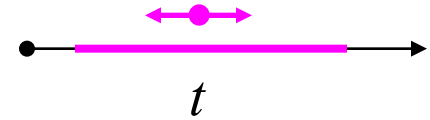


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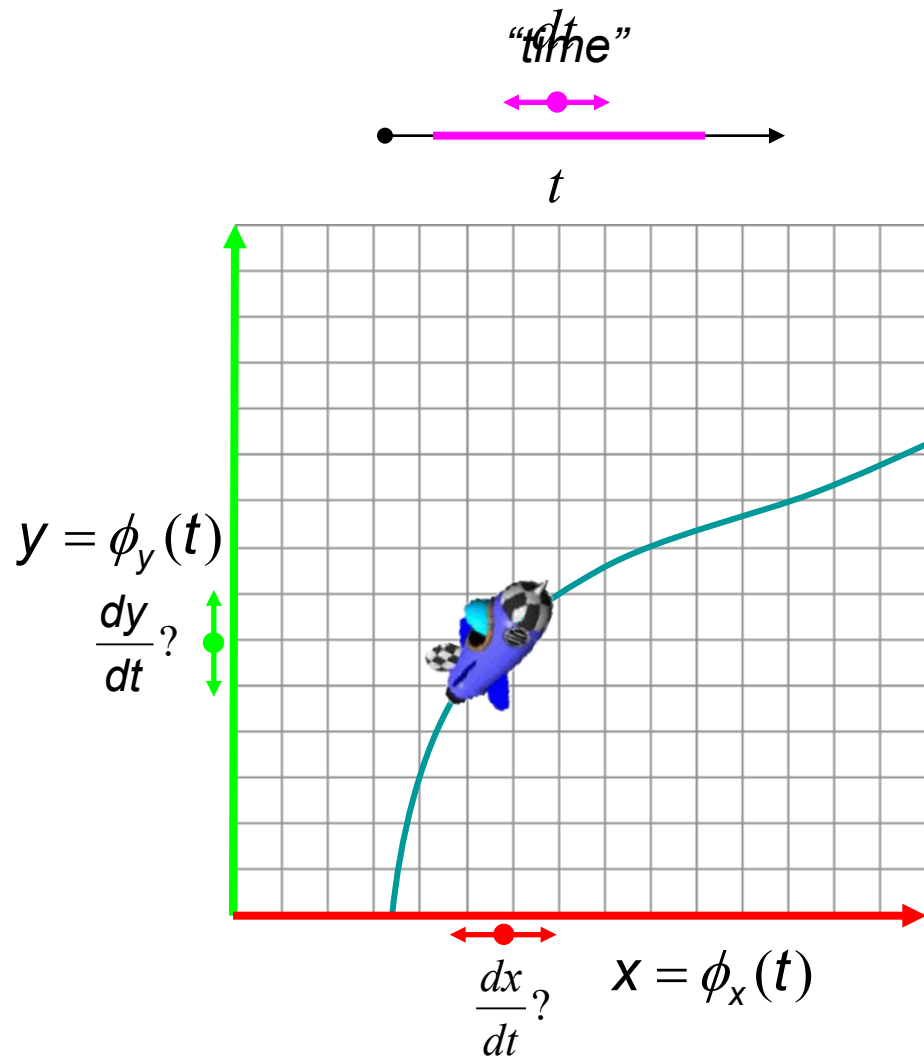


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# Parametric Curves

*“Driving a little car in time” analogy*



# Parametric Curves

Due to the **independence** of the basis functions  $x_i$ , (orthonormality)

The functions  $f_i$  can be seen as completely independent:

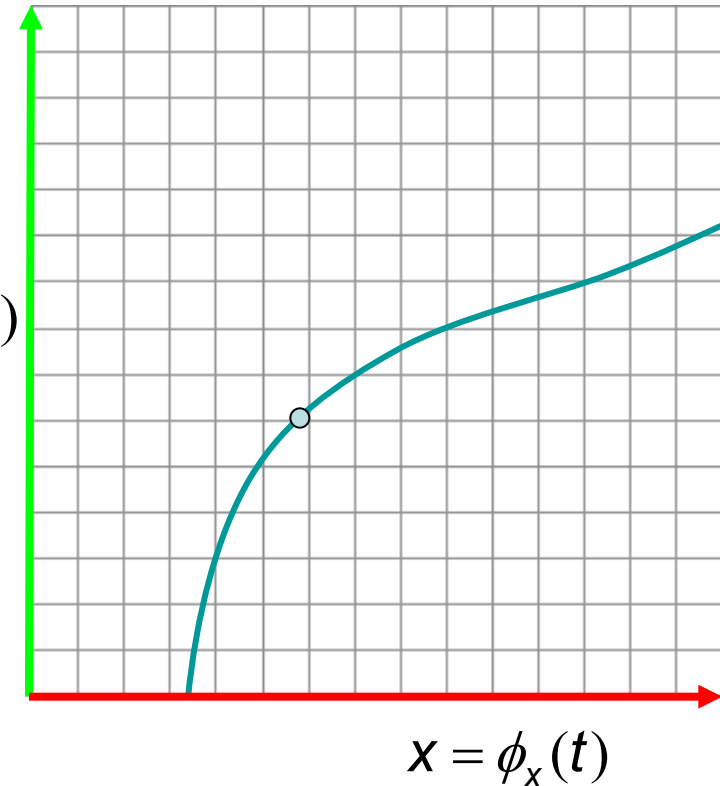
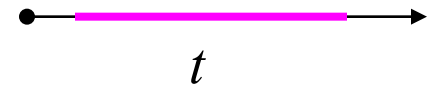
$$\begin{bmatrix} x = \phi_x(t) \\ y = \phi_y(t) \\ z = \phi_z(t) \end{bmatrix}$$

$$\mathbf{x}_i = \phi_i(t)$$

$$\mathbf{x} = \phi(t)$$

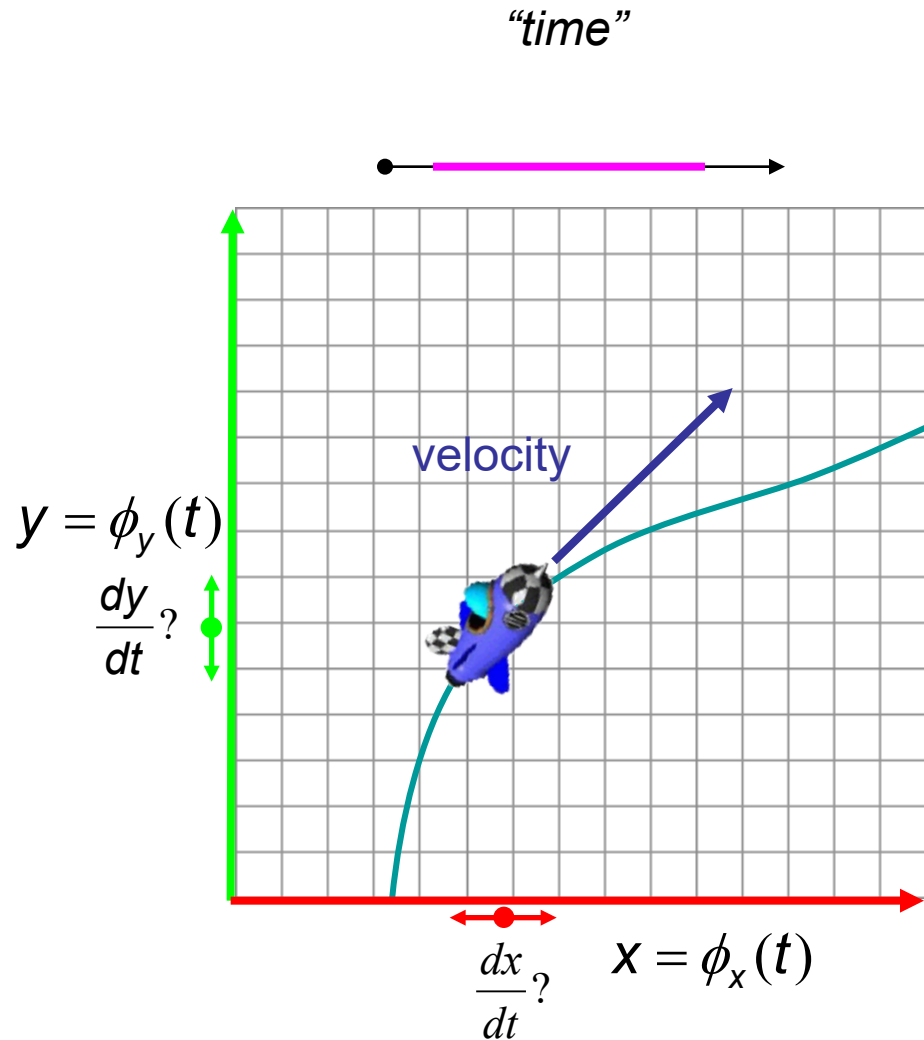
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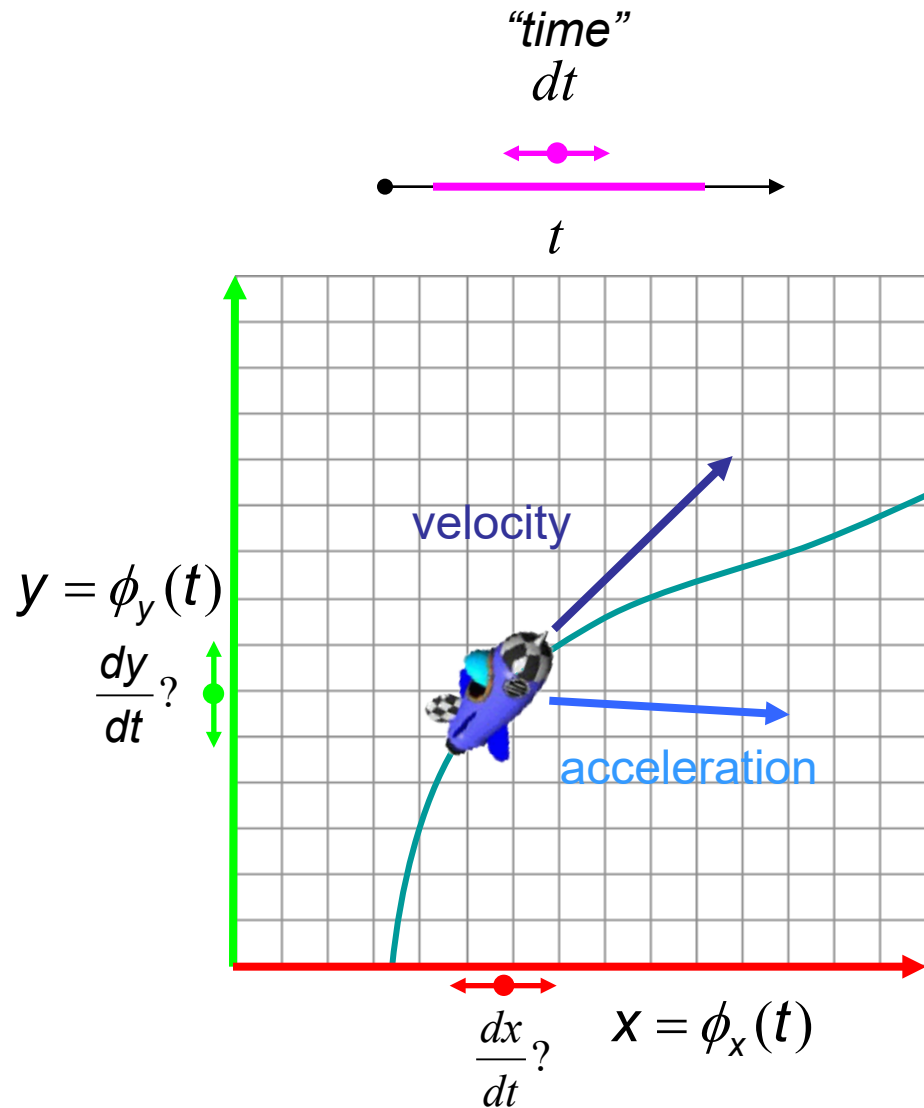
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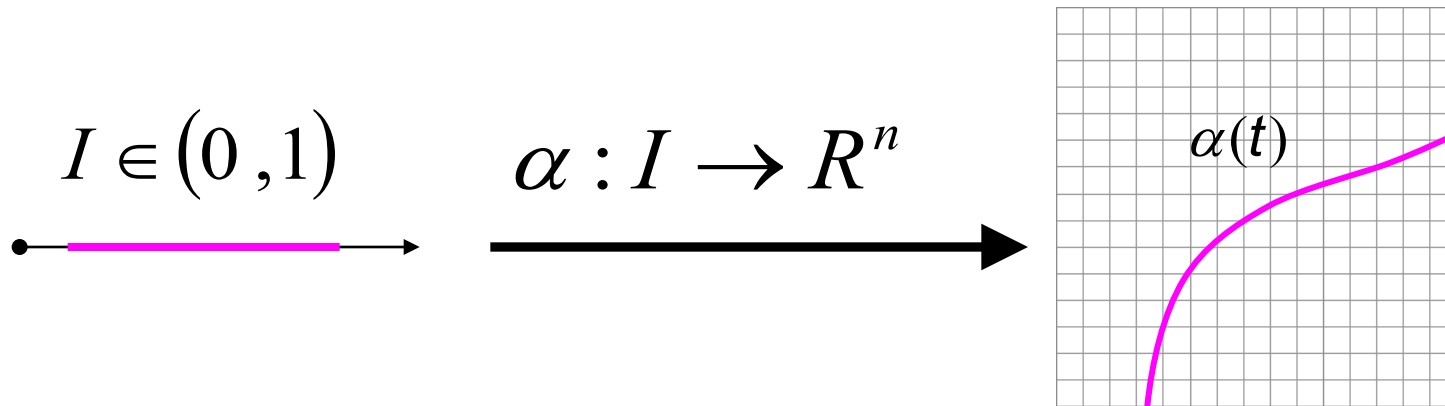


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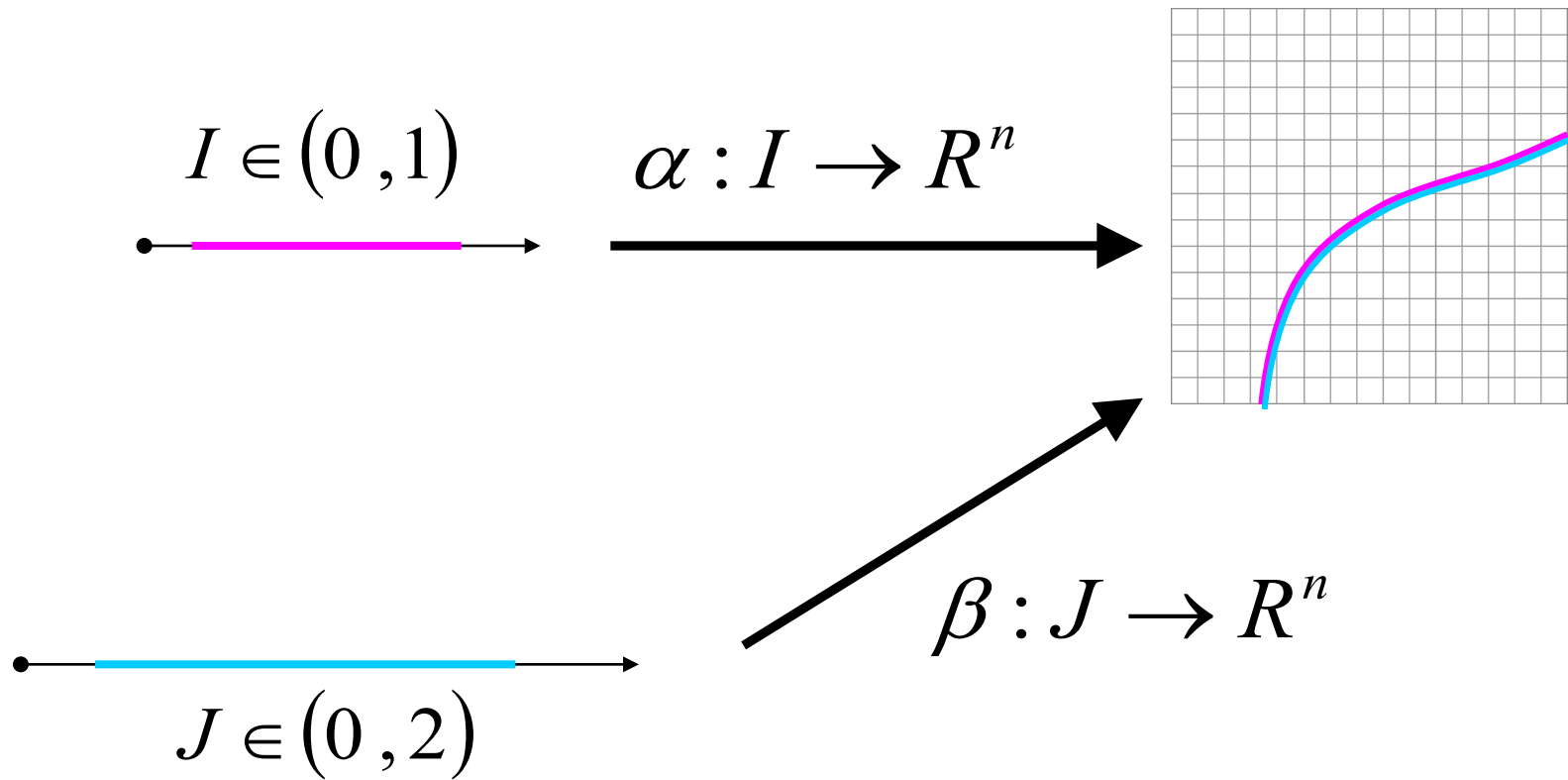


# Reparameterization



A path in “time” (or some other independent parameter)

# Reparameterization



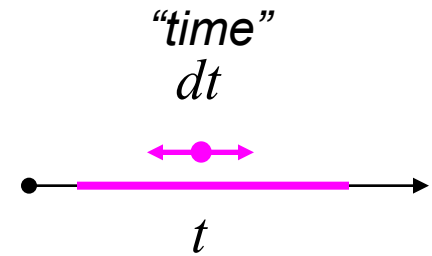
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*“Driving a little car in time” analogy*

We can imagine a  
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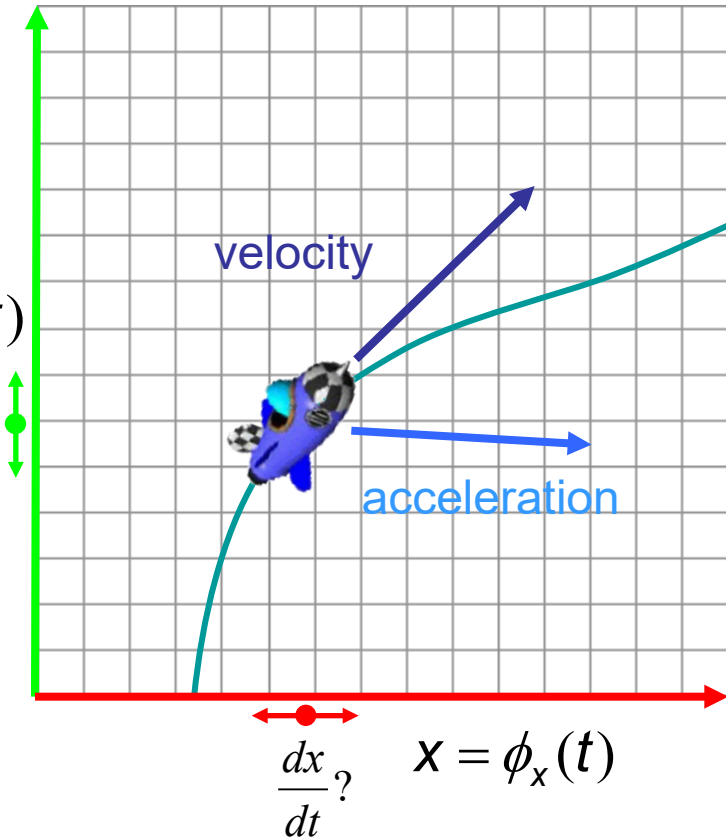
Where **t** is distance.

What happens??



$$y = \phi_y(t)$$

$$\frac{dy}{dt}?$$



# Parametric Curves

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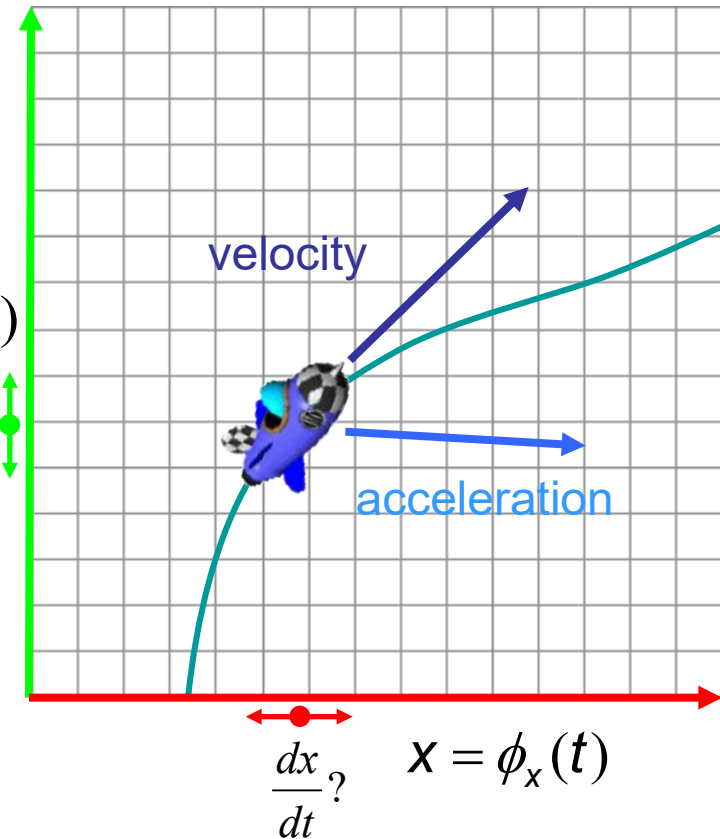
What happens??

$$\left\| \frac{d\mathbf{x}}{dt} \right\| = 1$$

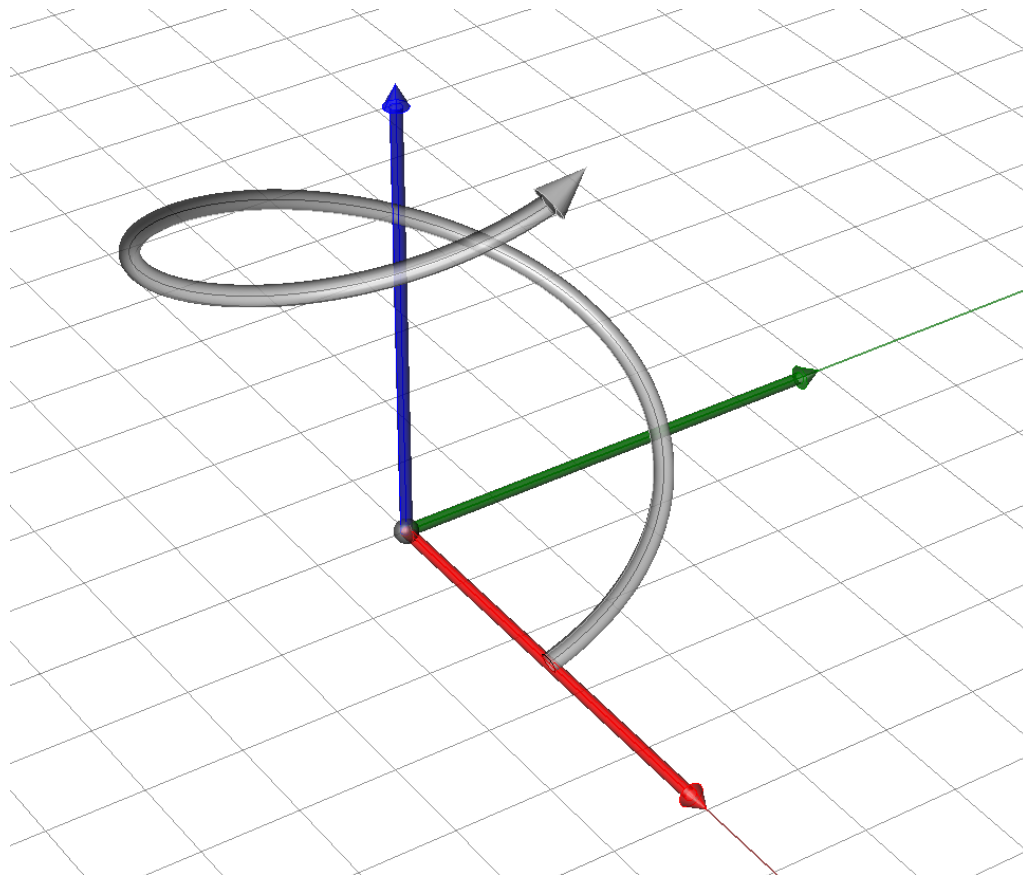
$$\left\| \frac{d^2\mathbf{x}}{dt^2} \right\| = \text{curvature}$$

$$y = \phi_y(t)$$

$$\frac{dy}{dt} ?$$

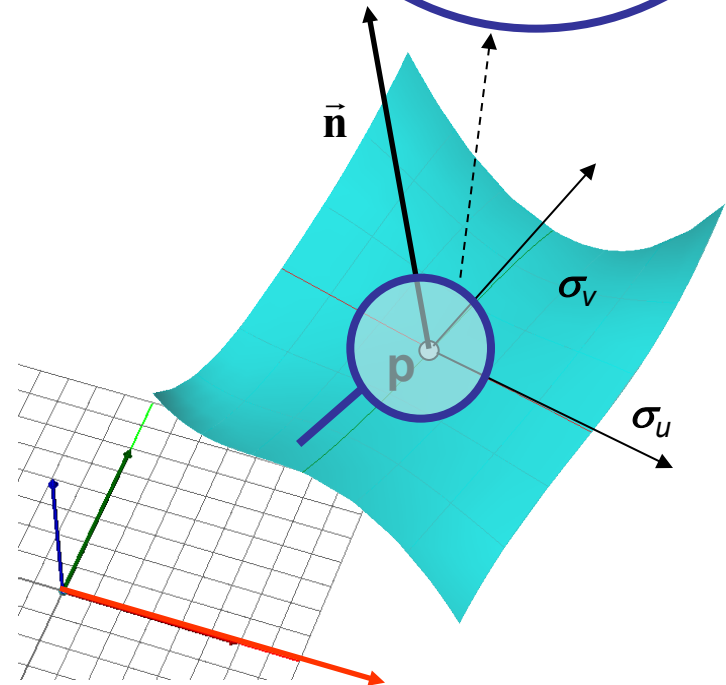
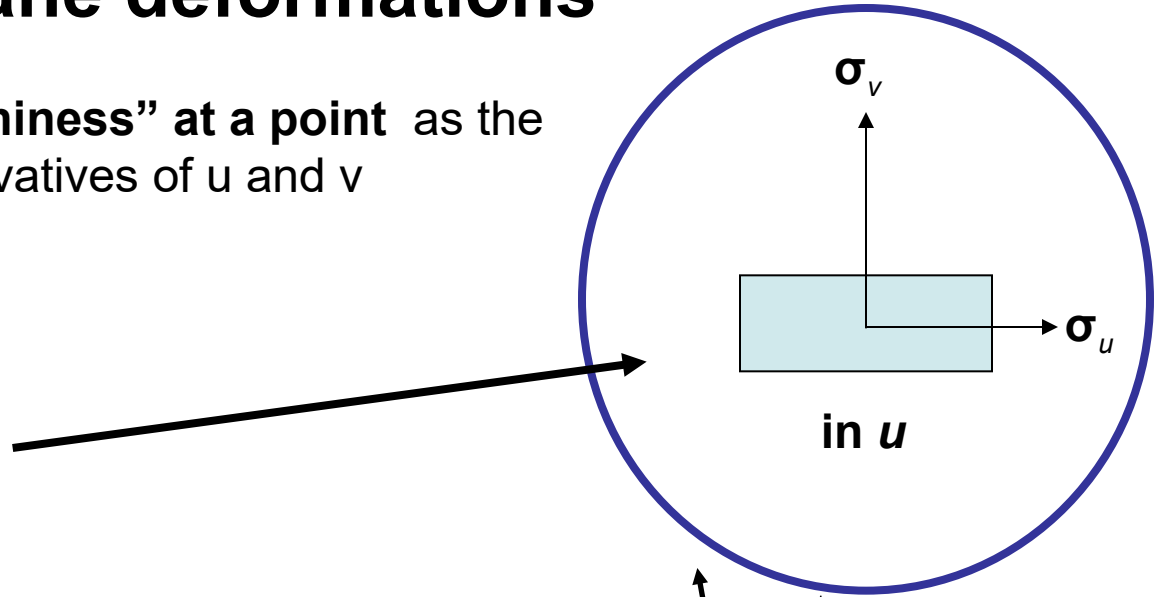
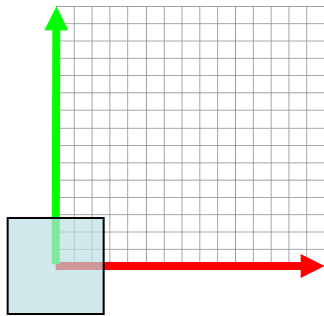


## ***Classic Case: The “unit speed” Helix in 3D***



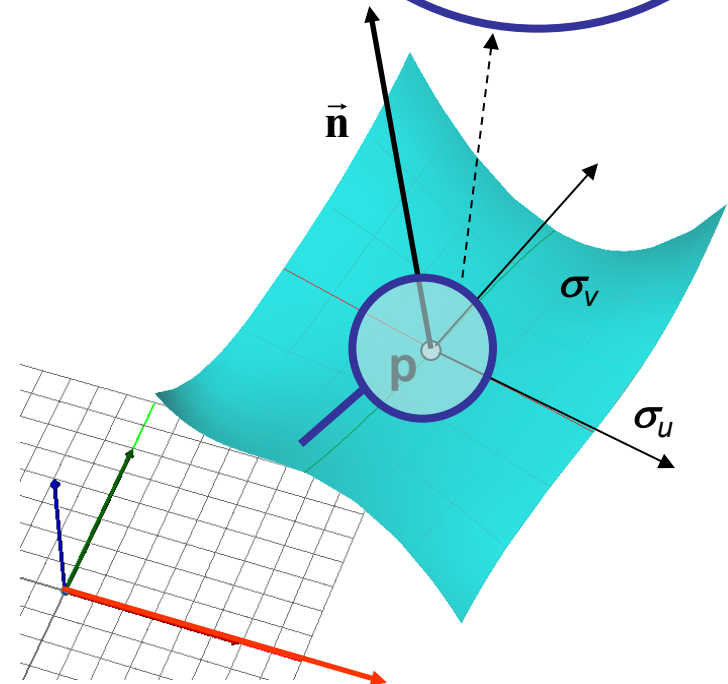
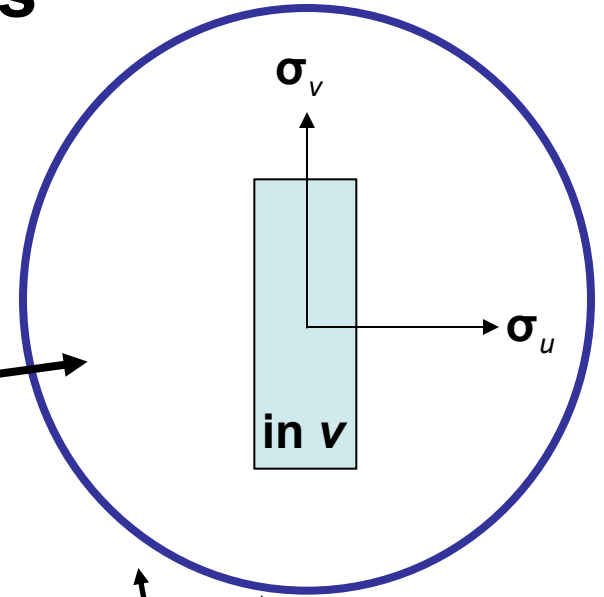
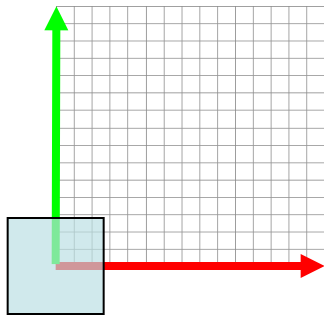
# First form: in plane deformations

We talked about “stretchiness” at a point as the magnitude of the first derivatives of  $u$  and  $v$



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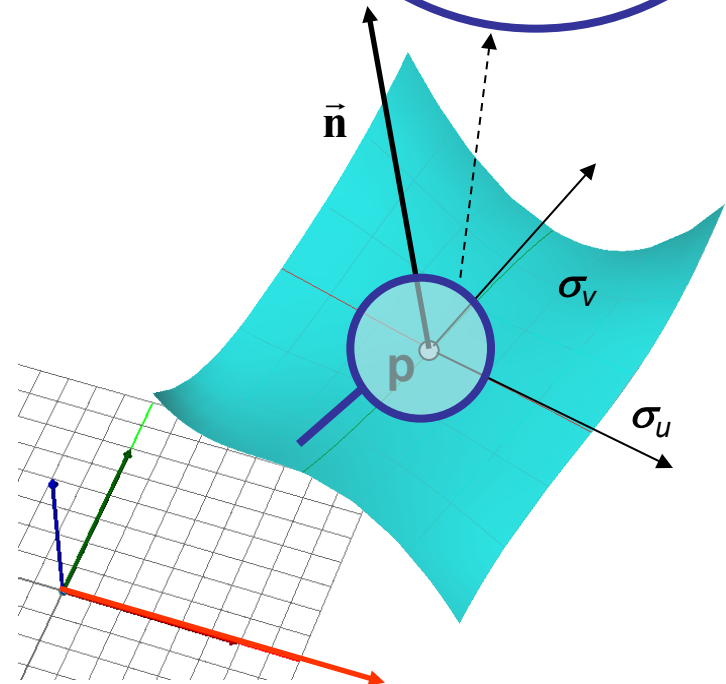
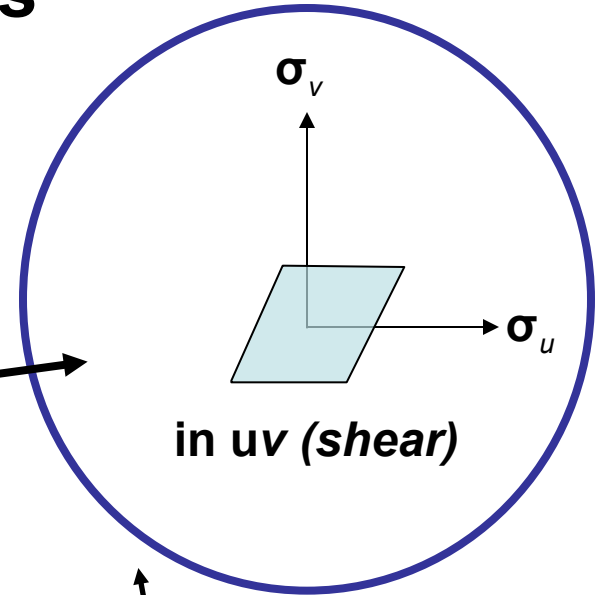
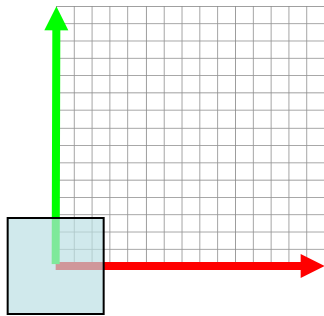
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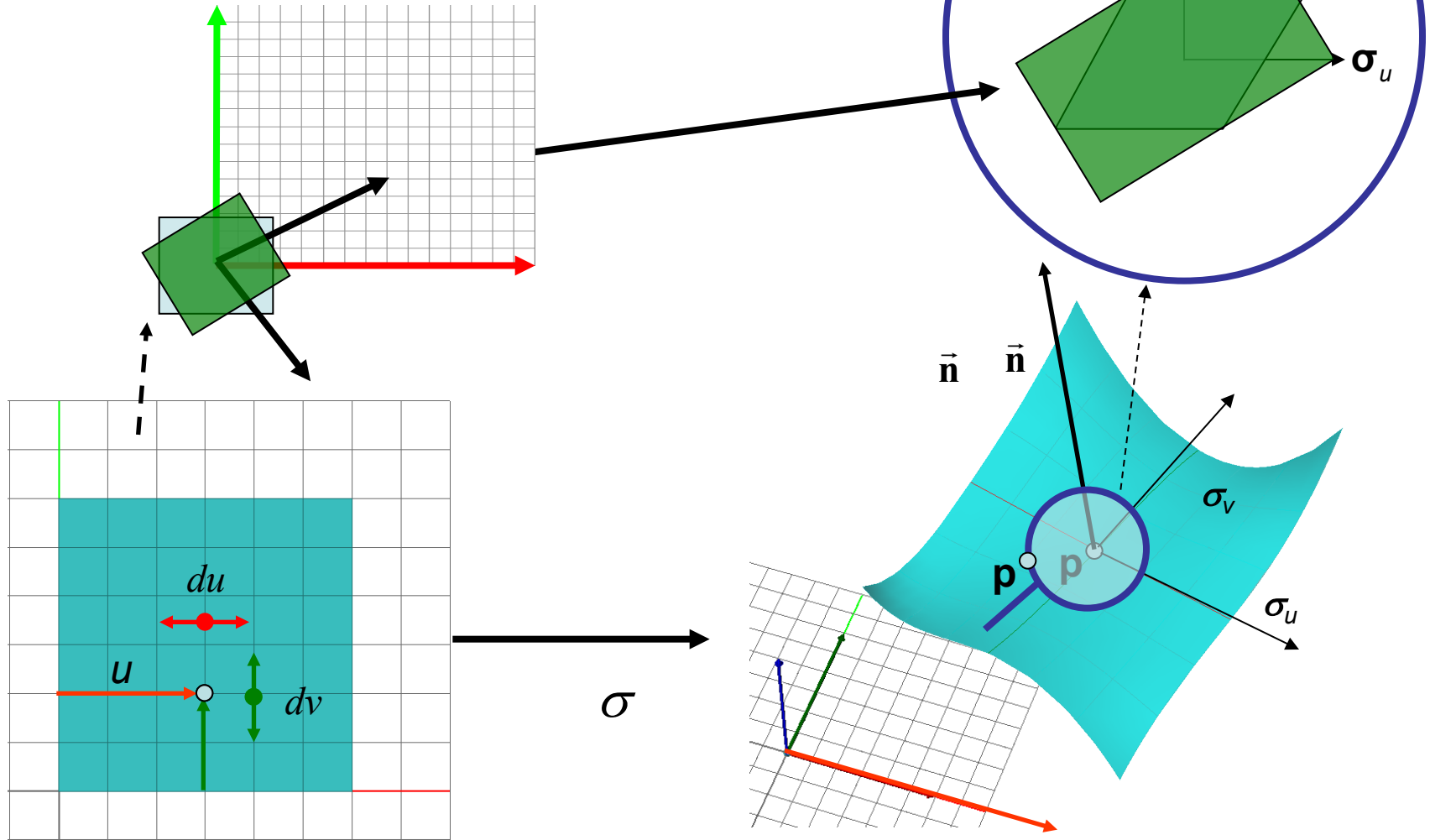
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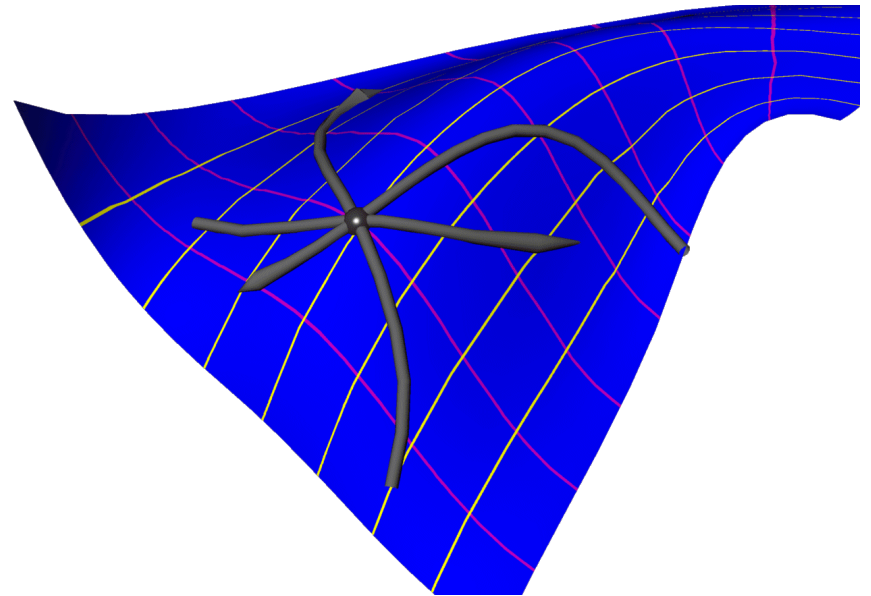
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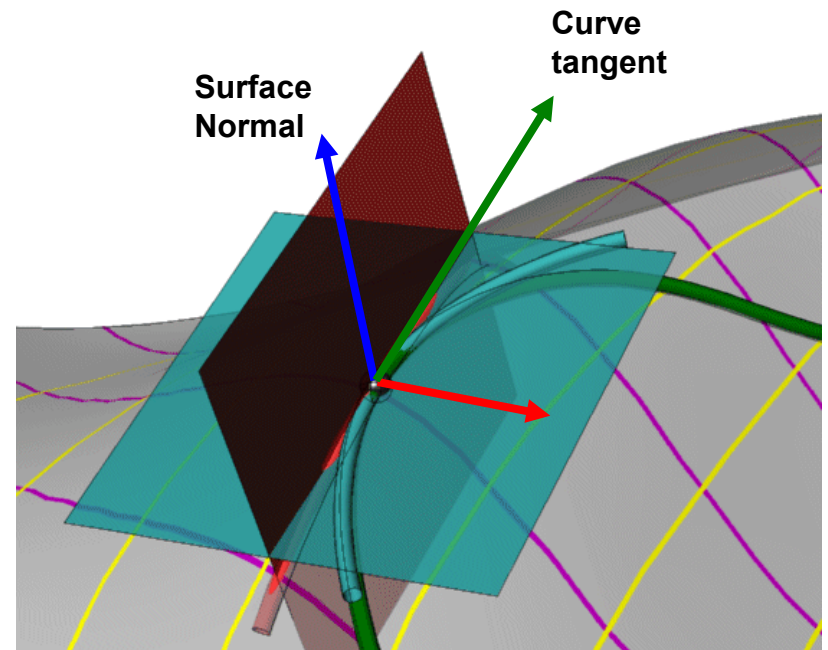
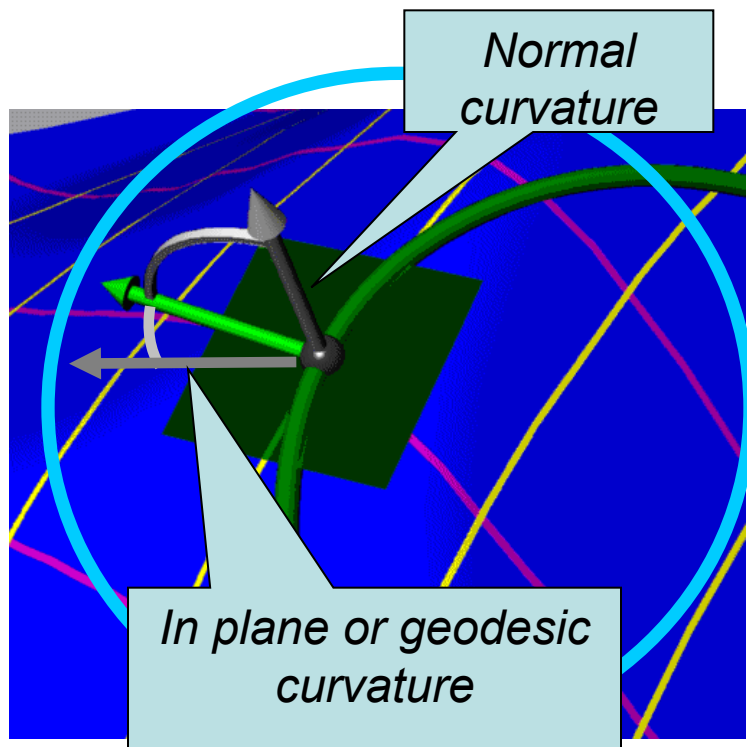
# Second fundamental form

We did the same thing for all possible curvatures  
(2<sup>nd</sup> derivatives)  
of the curves of a surface...



# Second fundamental form

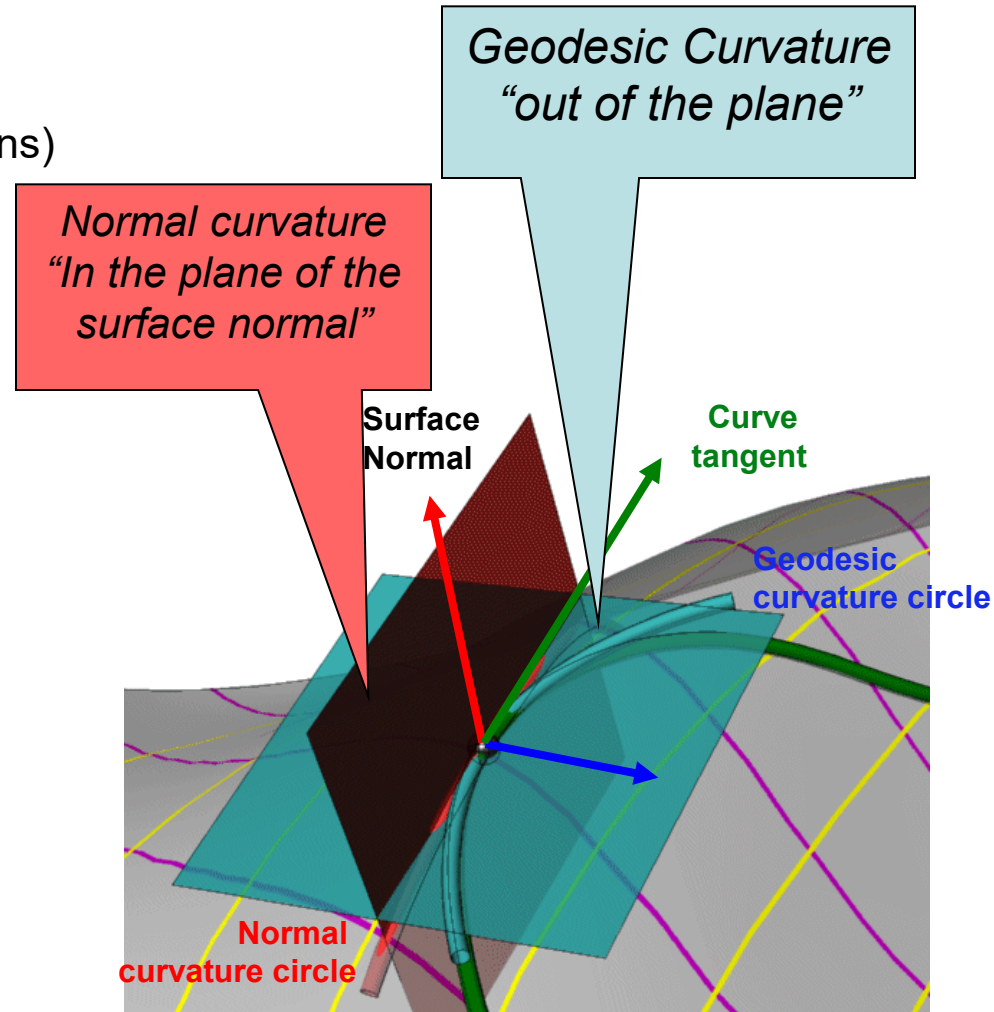
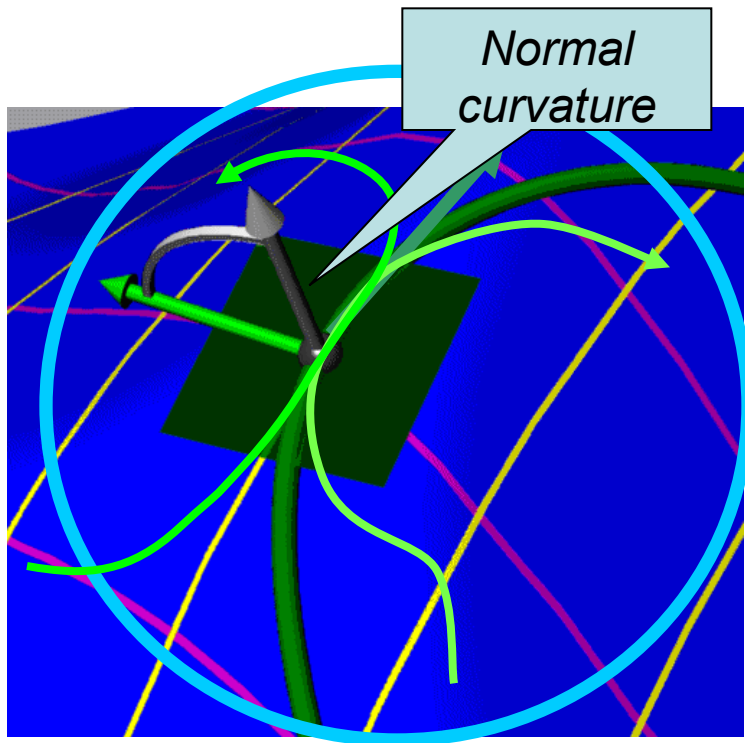
If we look at this from the frame defined by the curve tangent and the surface normal (and the vector perpendicular to both):



# Second fundamental form

Different surface curves with a given tangent direction have different geodesic curvatures, But the same normal curvature.

The normal curvature (in a given directions) of a curve on the surface is a feature of the surface itself.

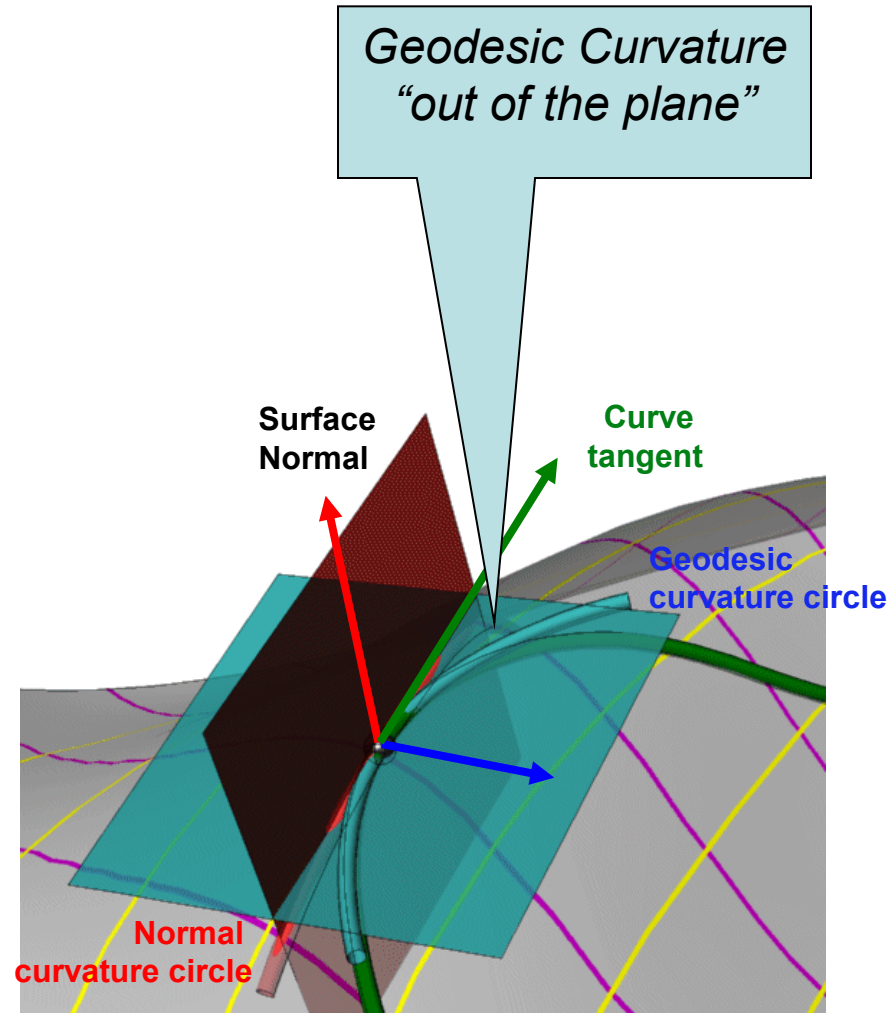
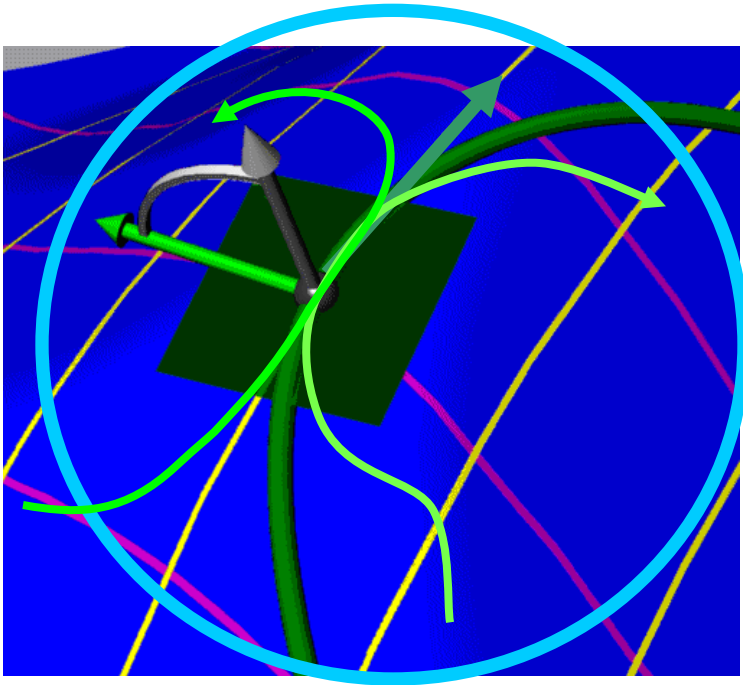


## Second fundamental form

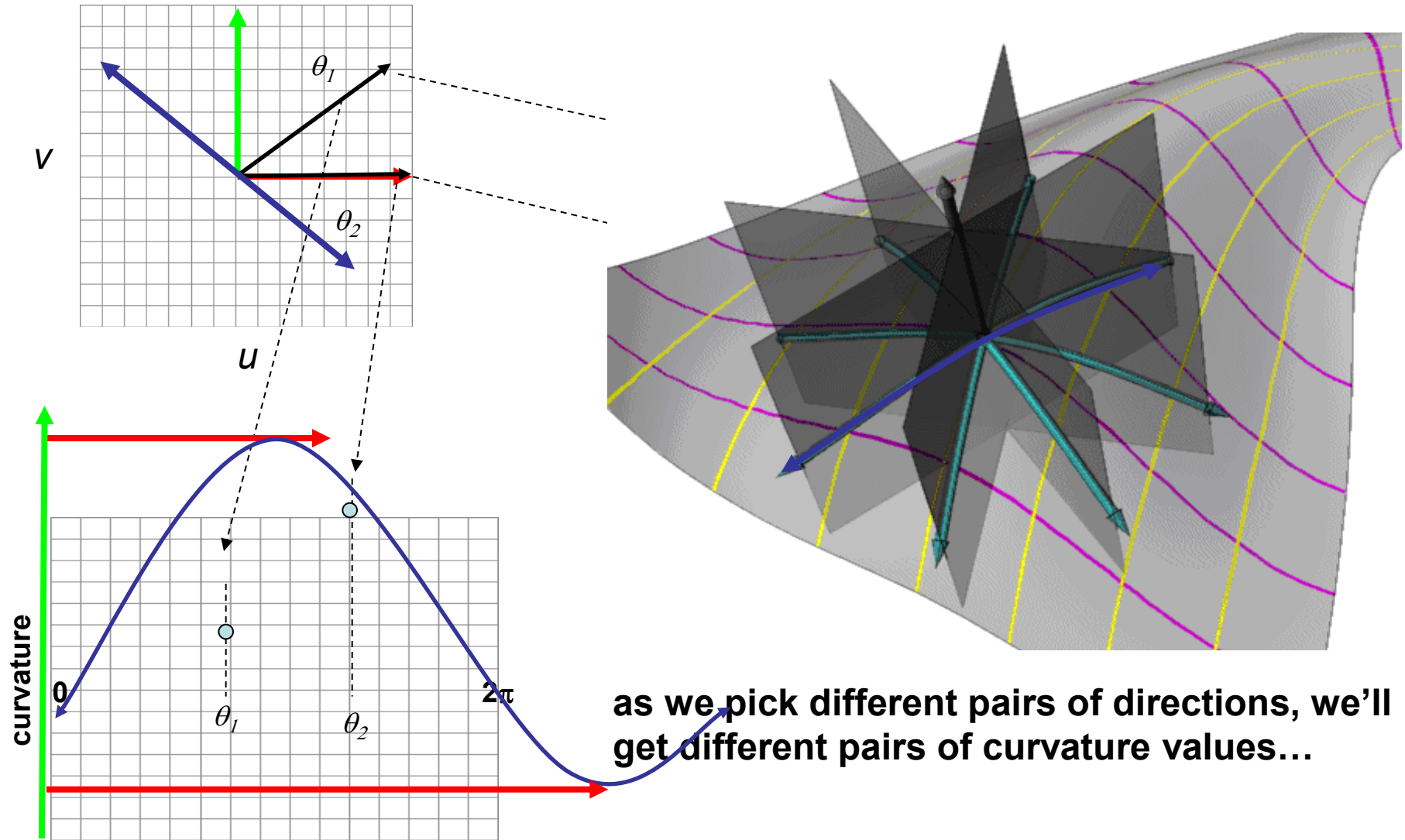
$$\hat{\mathbf{n}} \bullet \boldsymbol{\sigma}_{uu}, \hat{\mathbf{n}} \bullet \boldsymbol{\sigma}_{uv}, \hat{\mathbf{n}} \bullet \boldsymbol{\sigma}_{vv}$$

Different surface curves with a given tangent direction have different geodesic curvatures, But the same normal curvature.

The normal curvature (in a given directions) of a curve on the surface is a feature of the surface itself.

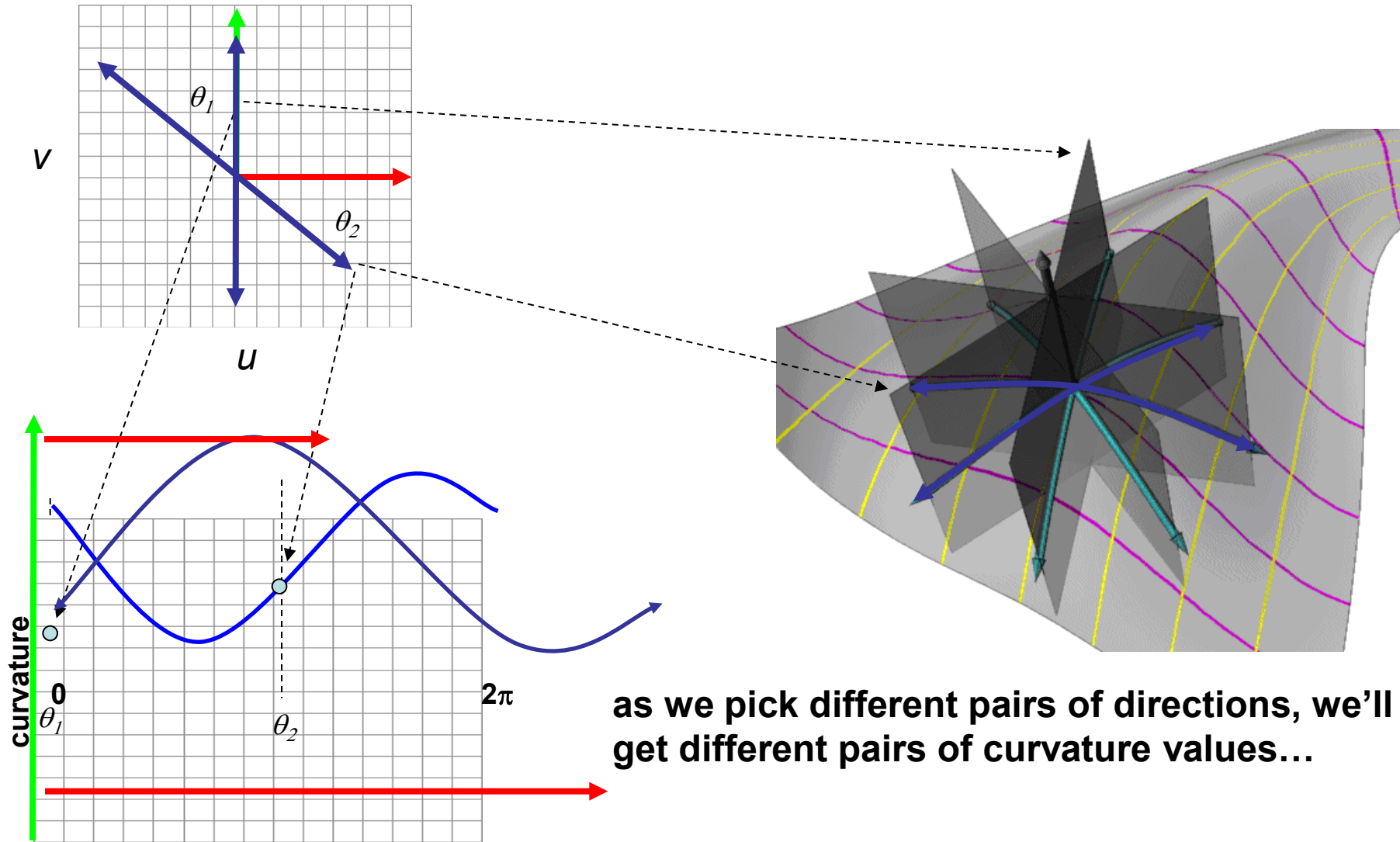


# Principle directions





# Bi- directions

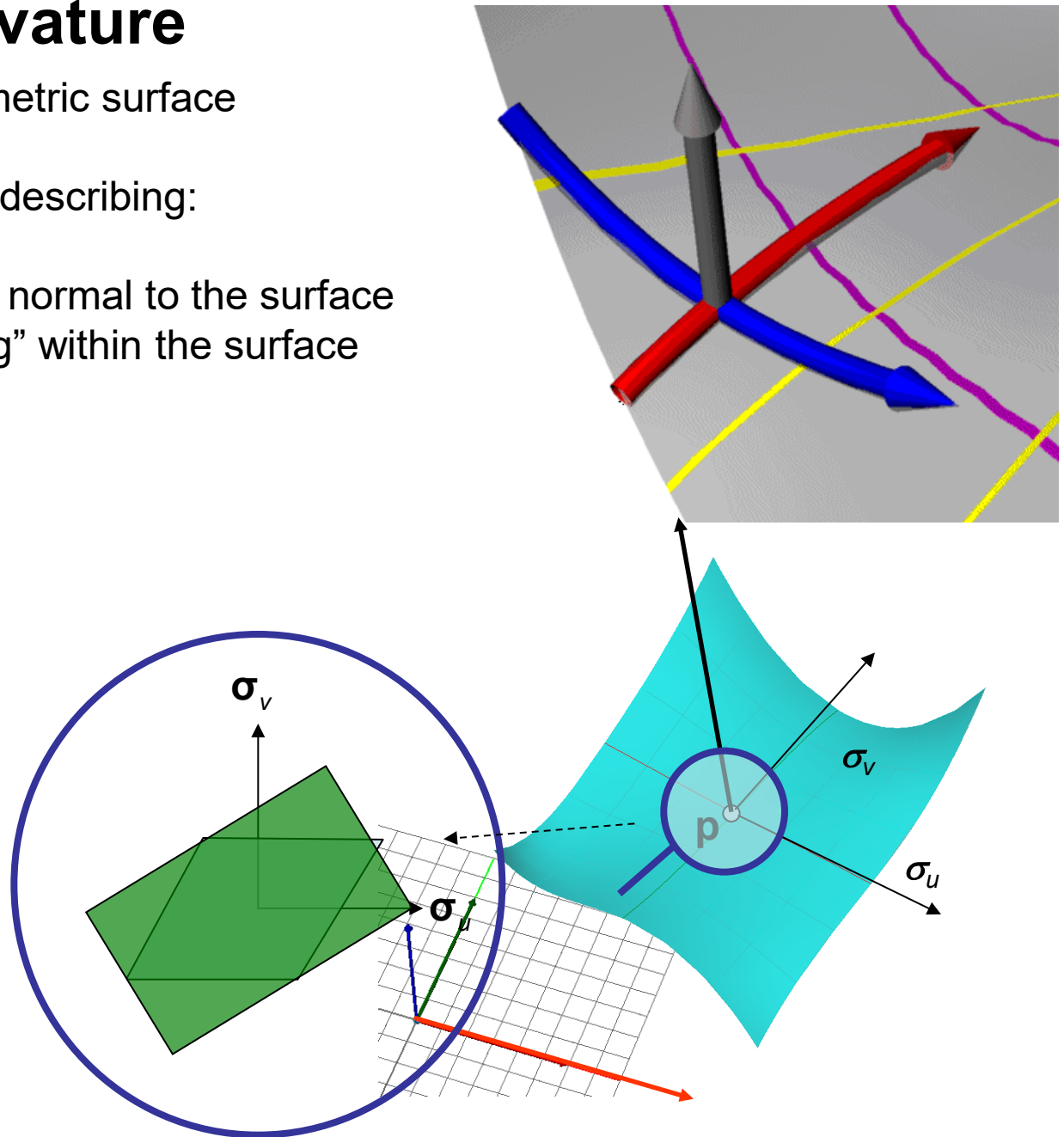




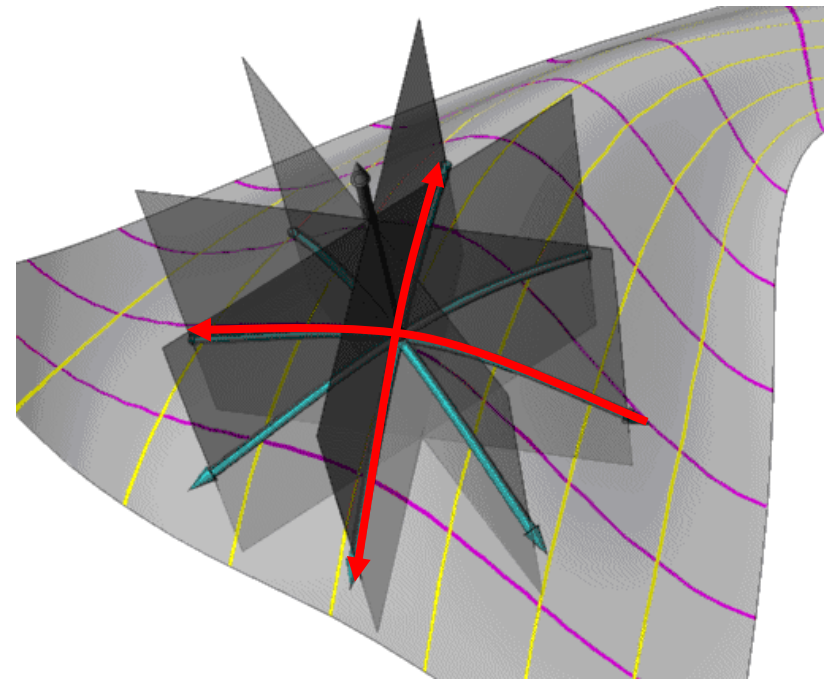
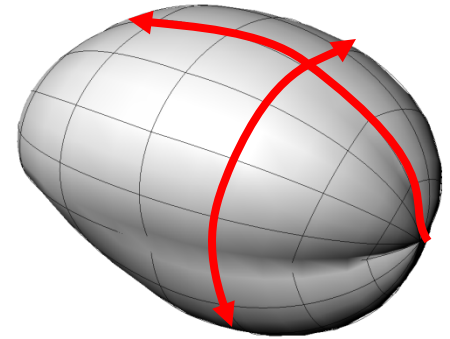
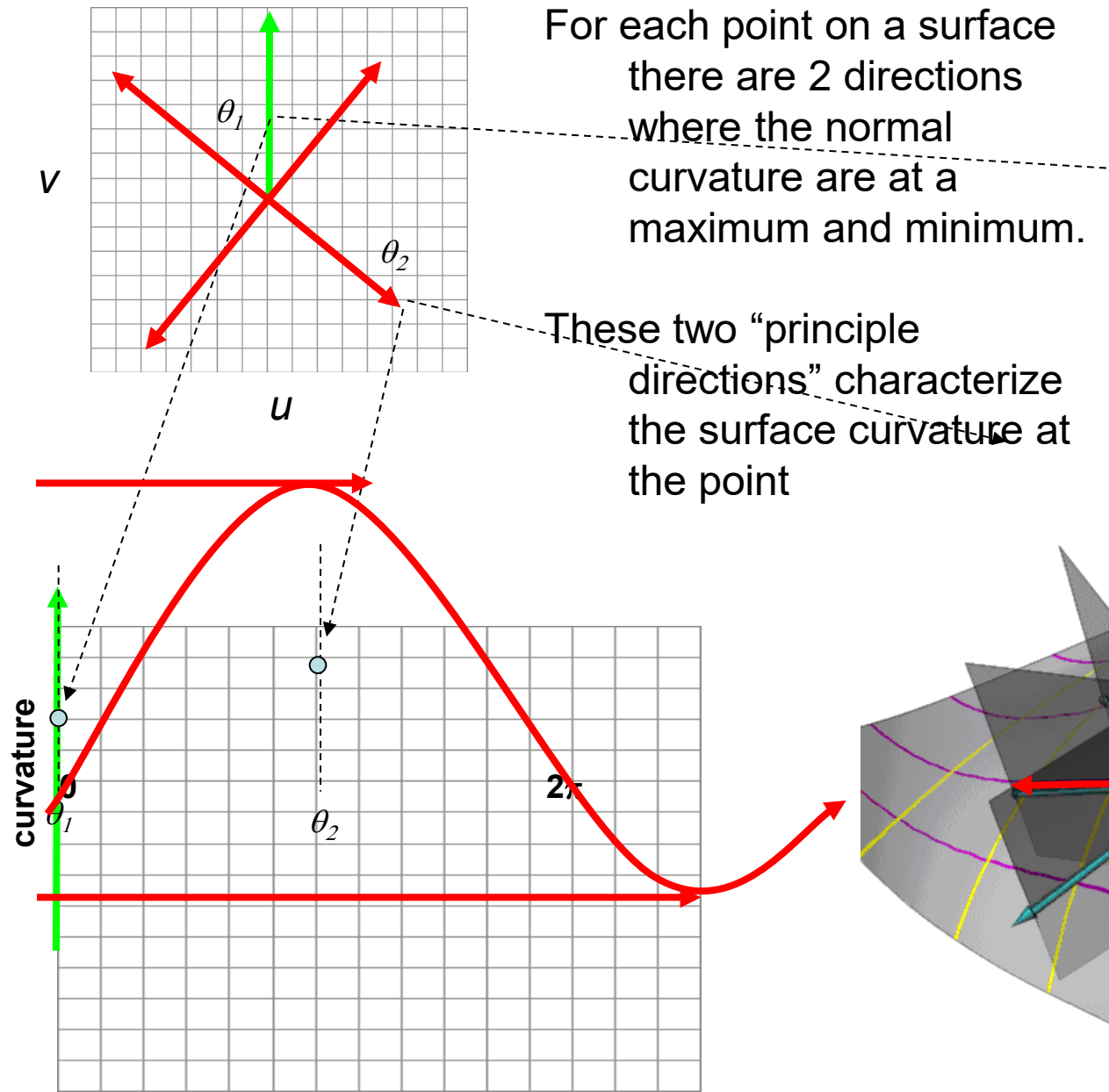
# Gaussian curvature

We know that a parametric surface is characterized by “bivectors” (vectors2) describing:

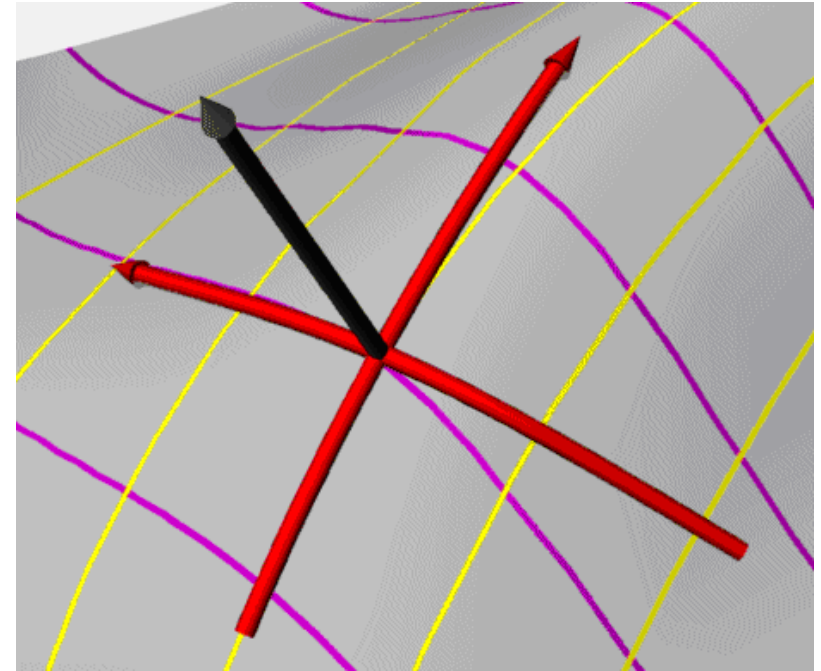
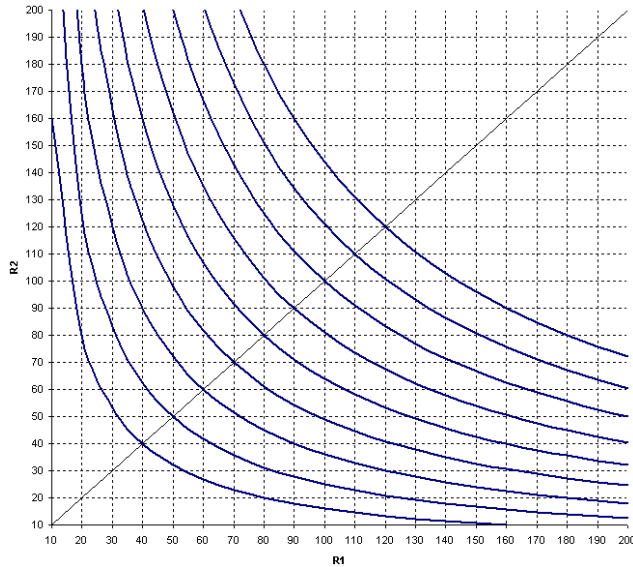
- It’s “acceleration” normal to the surface
- And its “stretching” within the surface



# Principle directions



# Gaussian Curvature: Definition and meaning #1

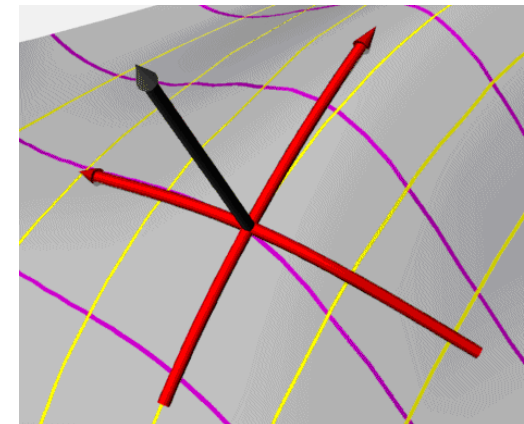
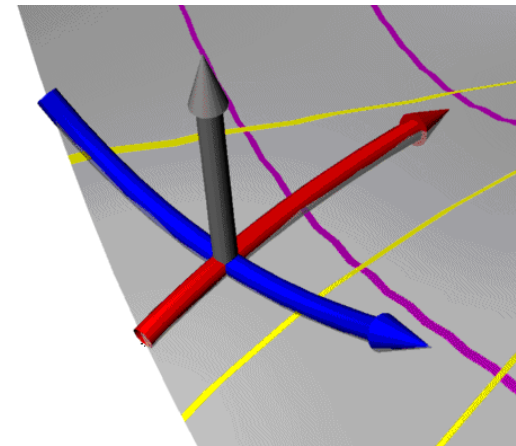
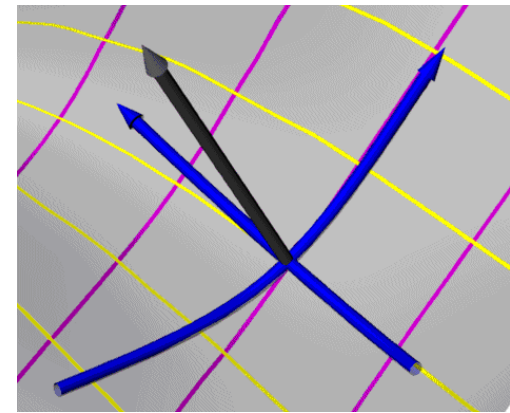
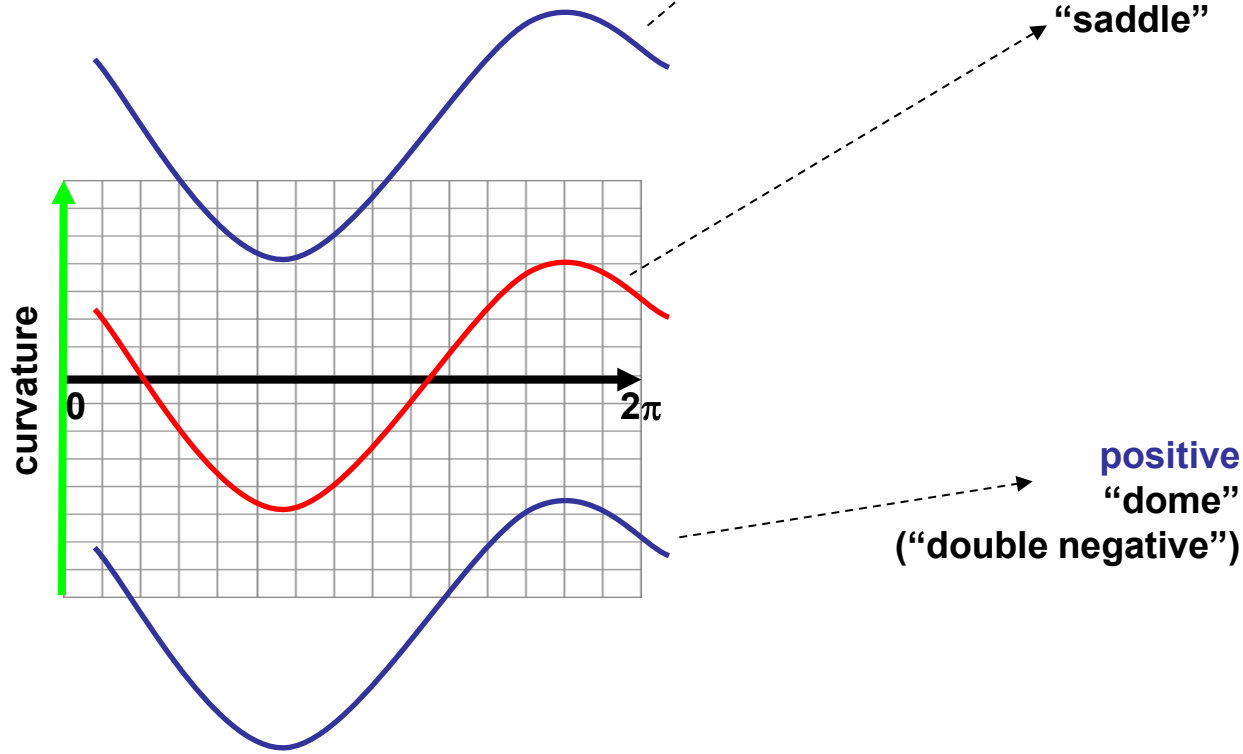


Q: What is  
A: The geometric mean, or  
“multiplicative average”

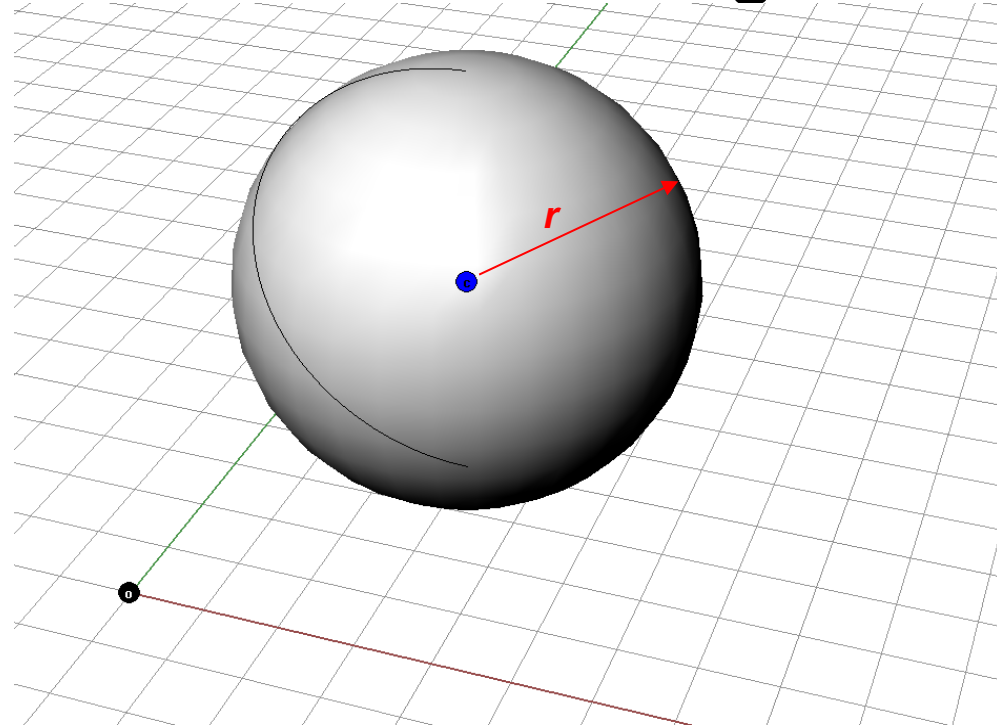
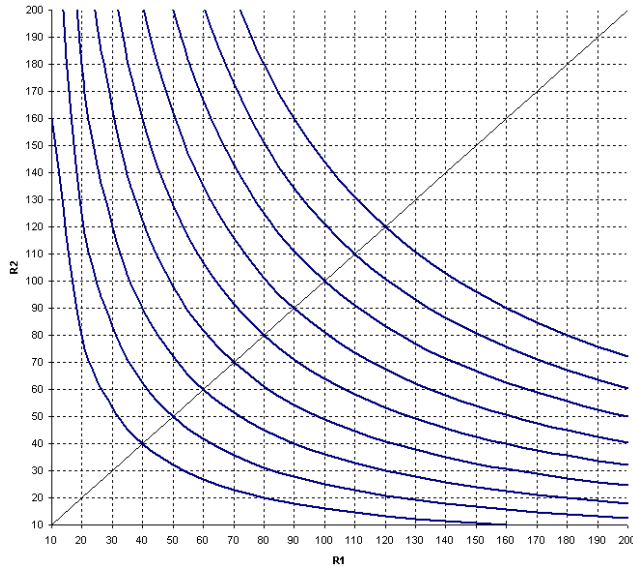
$$\sqrt{\frac{1}{\kappa}}$$
$$\sqrt{r_1 \times r_2}$$

# Curvature Sign characteristics

whether they curve in the **same direction** relative to the normal or in **opposing directions**.



# Gaussian Curvature: Definition and meaning #1

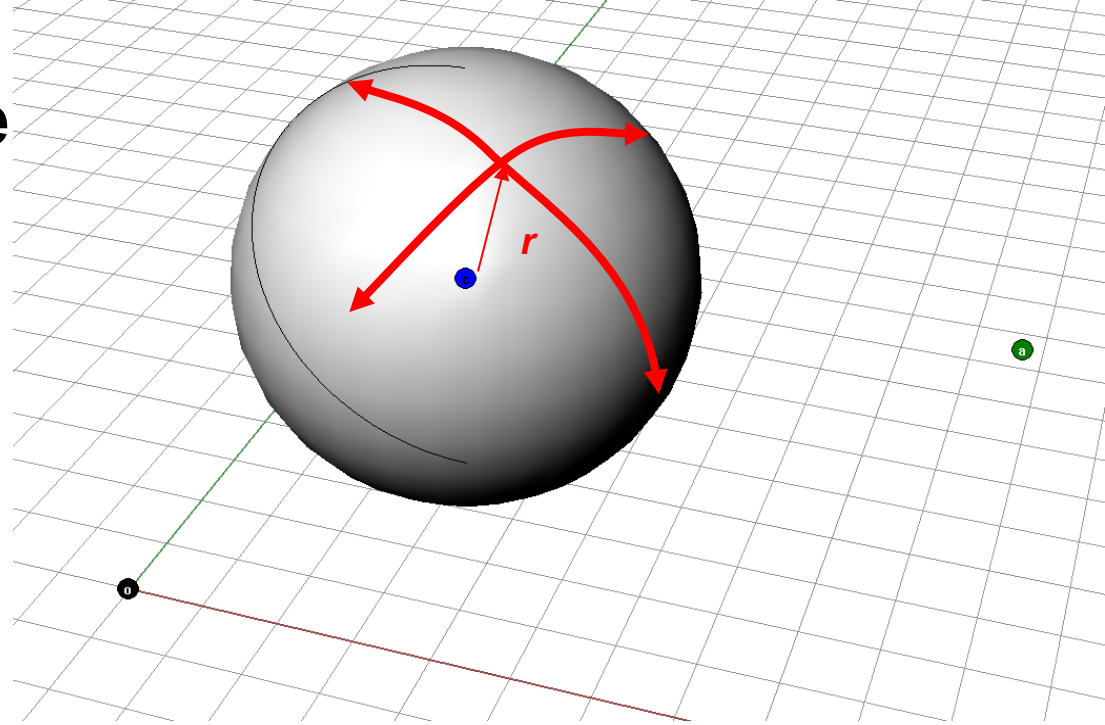
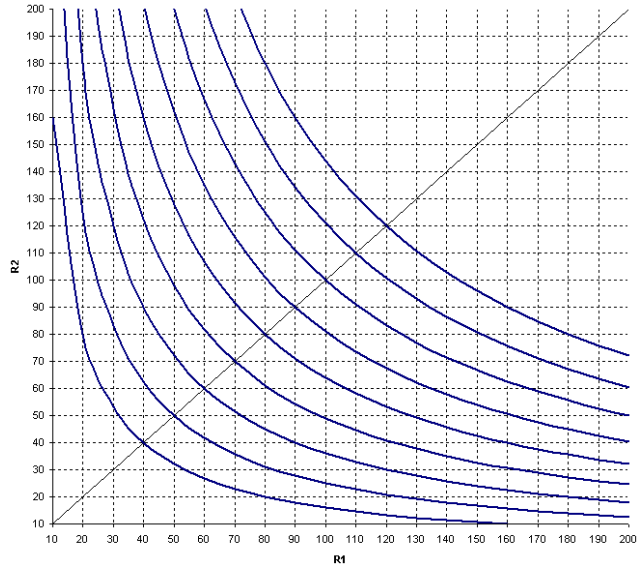


*Example: what are*  
*For a sphere of radius 10??*

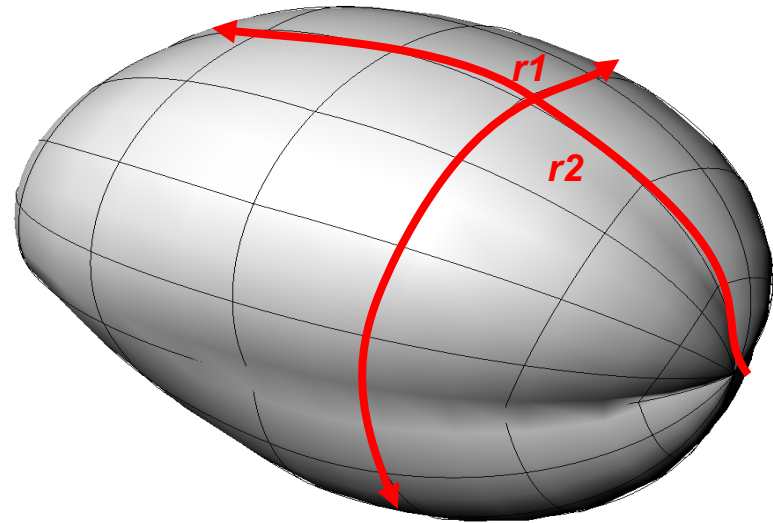
$$\begin{aligned} K &= c_1 \times c_2 \\ &= \frac{1}{r_1 \times r_2} \end{aligned}$$

$$\sqrt{\frac{1}{K}} = \sqrt{r_1 \times r_2}$$

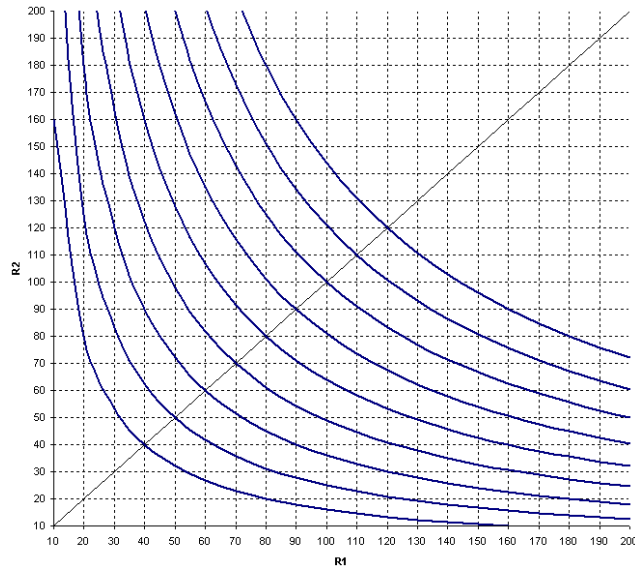
# Gaussian Curvature



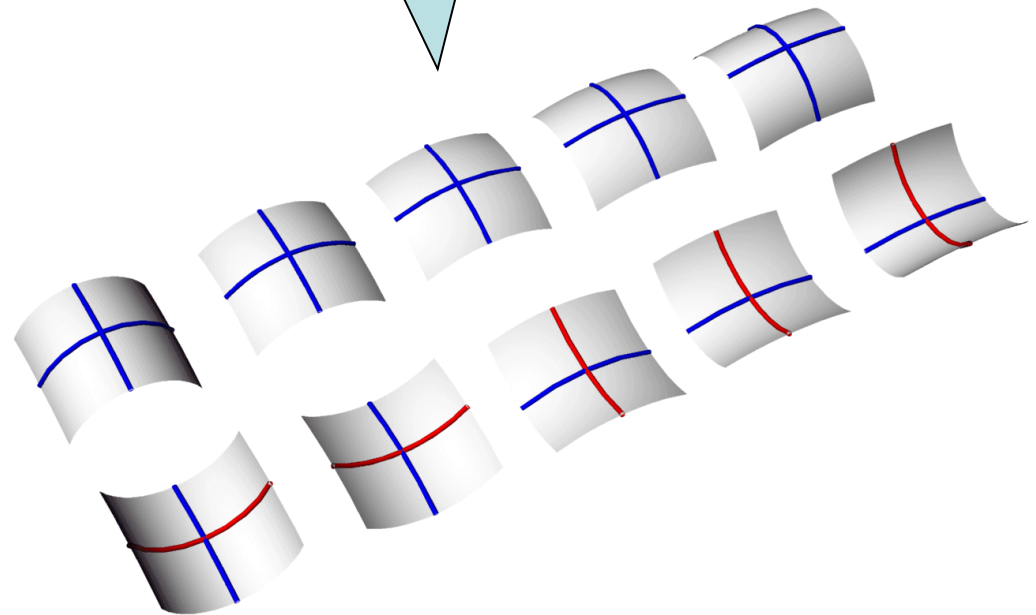
$$K = C_1 \times C_2$$
$$= \frac{1}{r_1 \times r_2}$$



# Gaussian Curvature: Definition and meaning #1

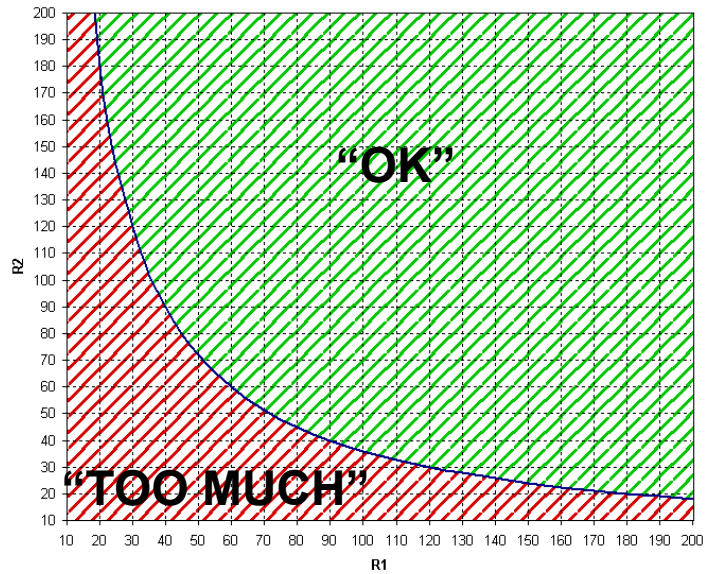
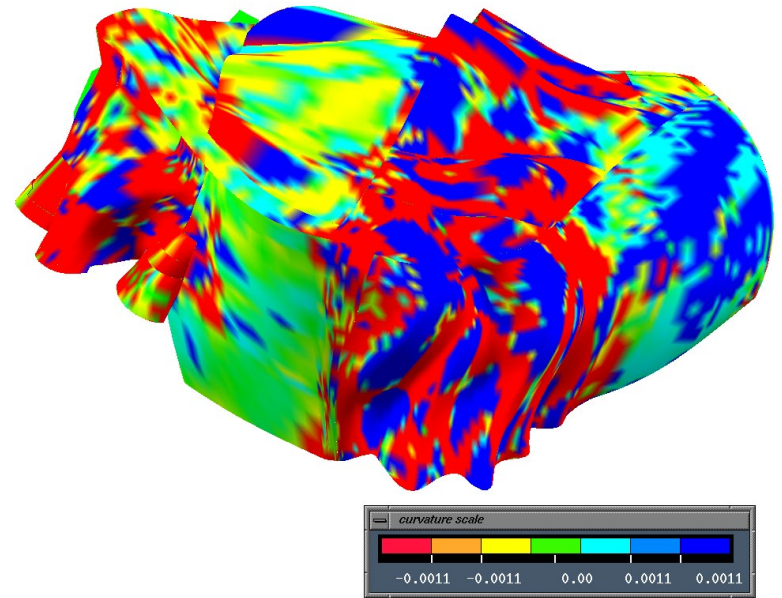
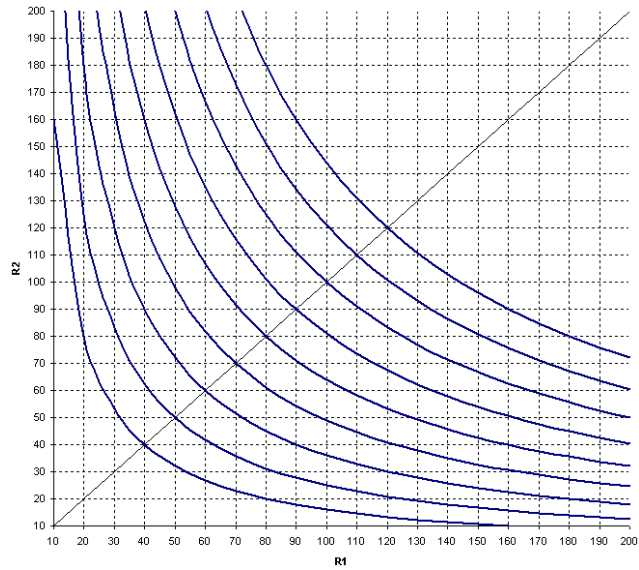


*Equivalent Gaussian curvatures*



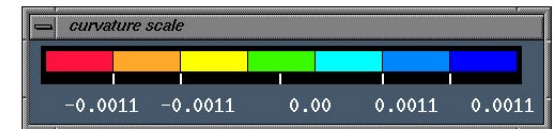
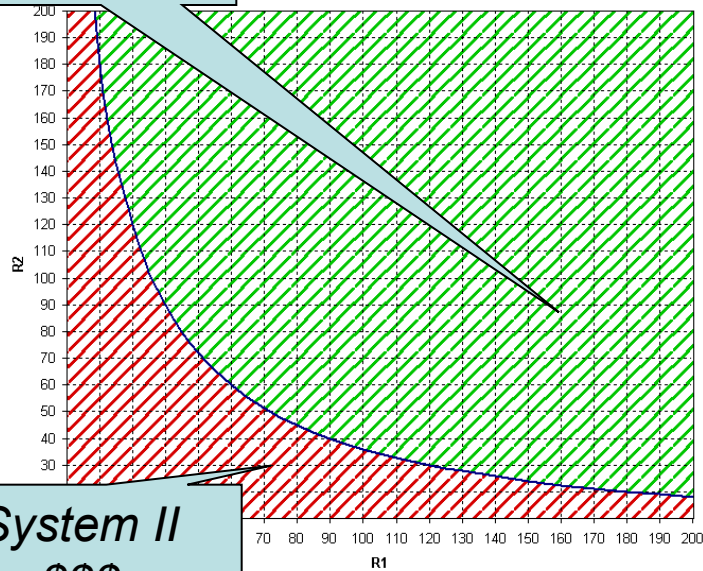
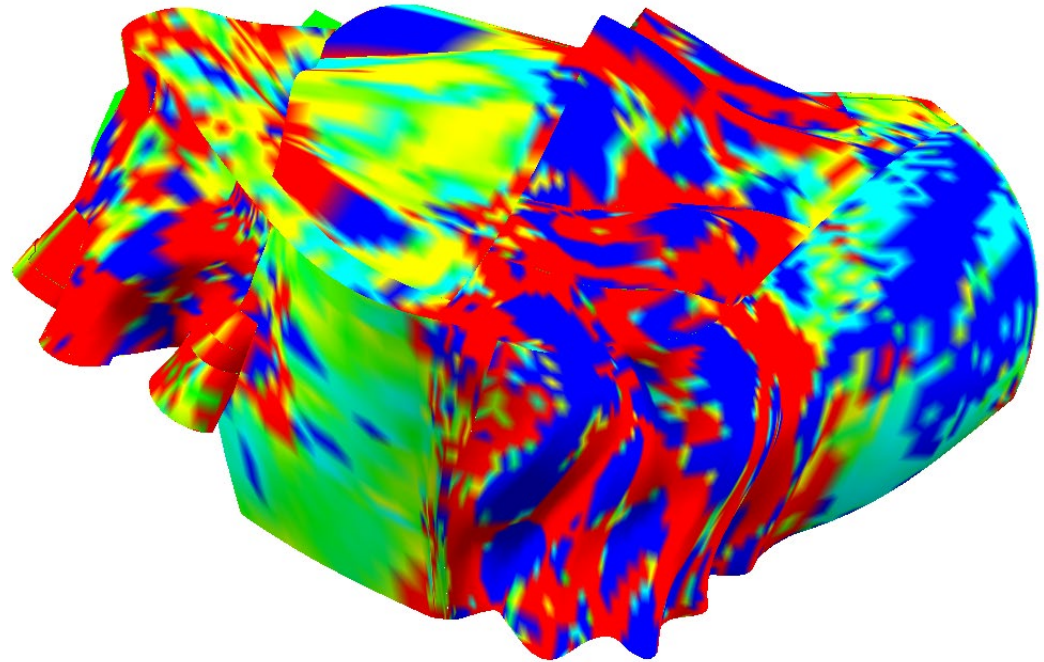
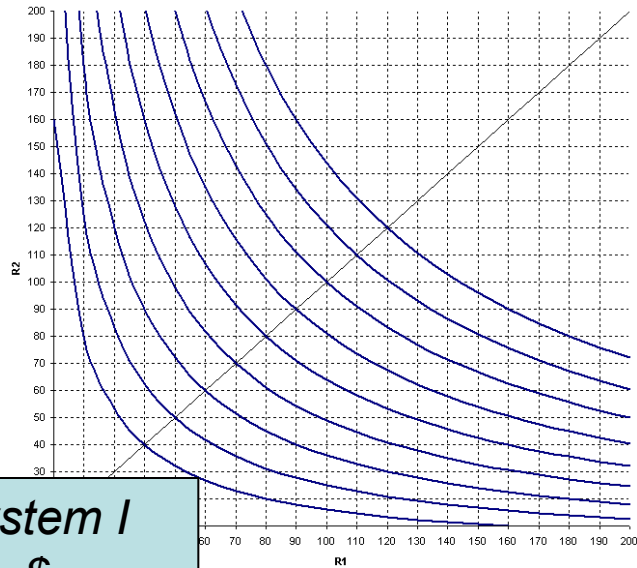


# Gaussian Curvature





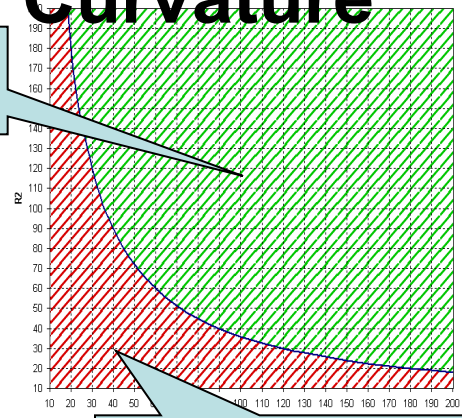
# Gaussian Curvature



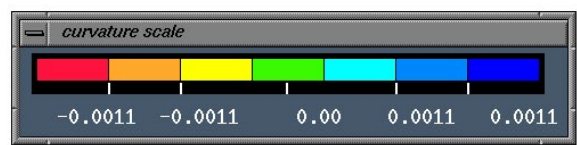
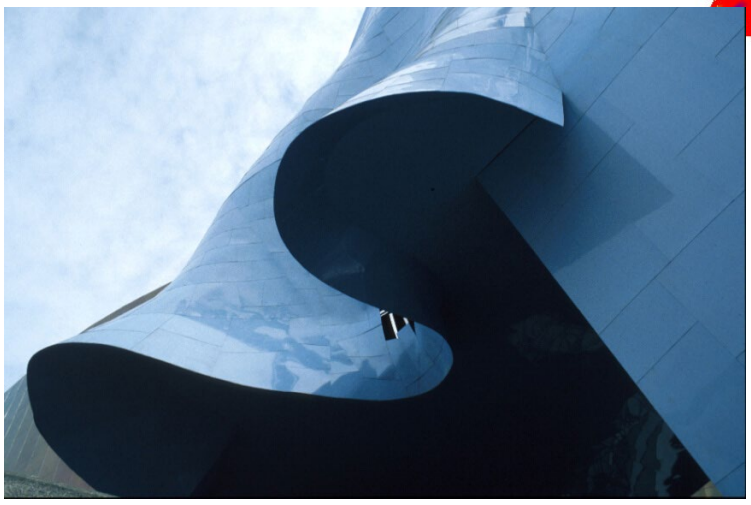
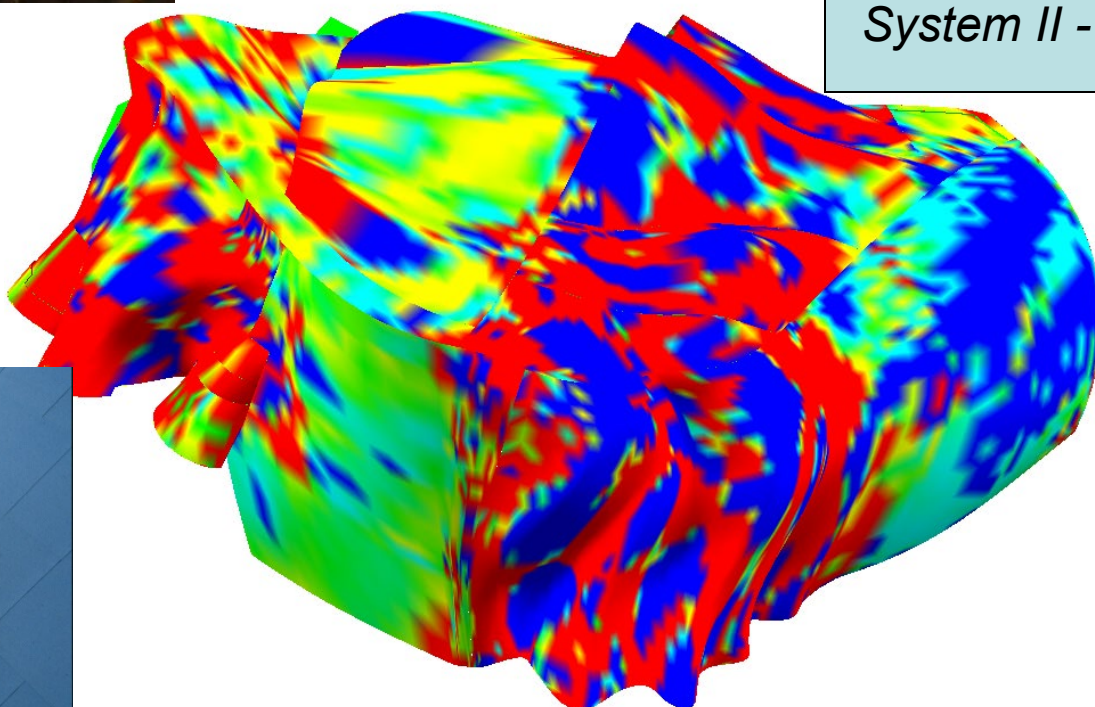


# Gaussian Curvature

System I - \$

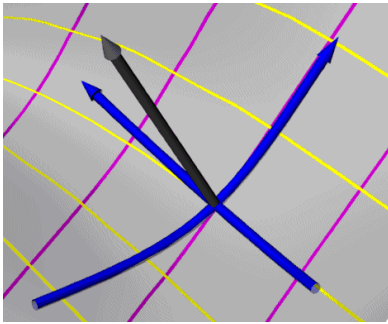


System II - \$\$\$



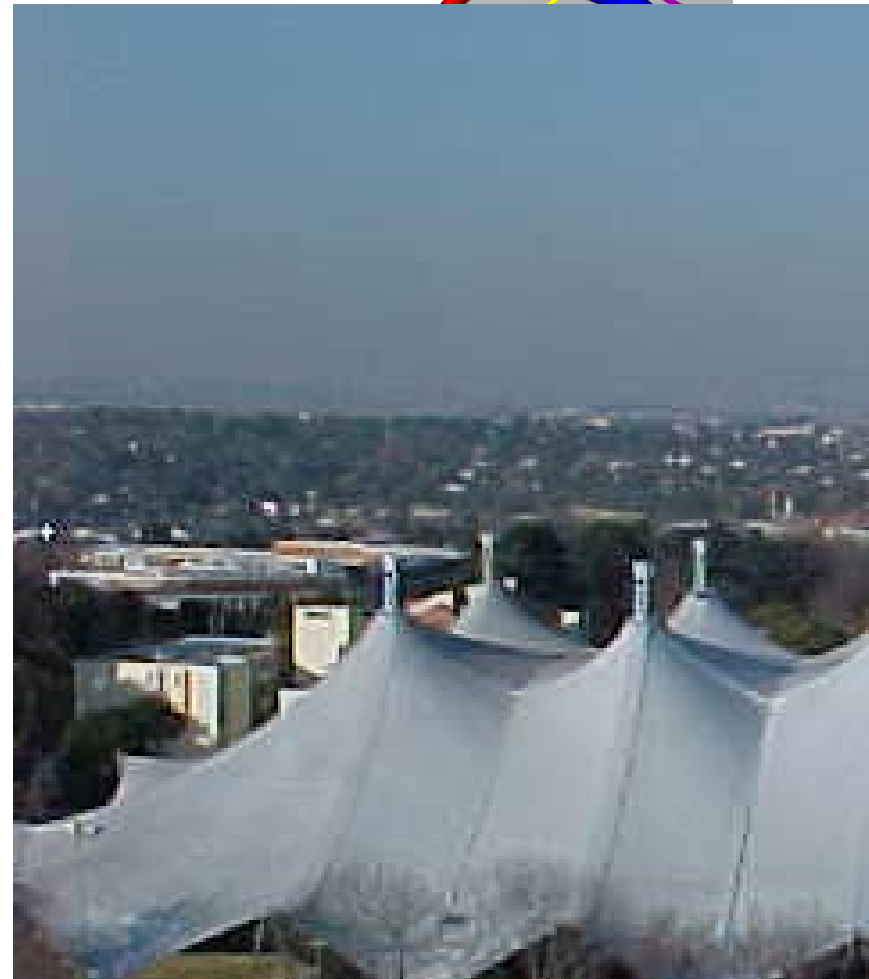
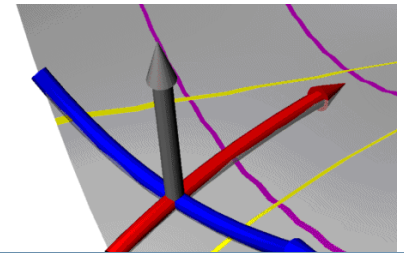


# Interpretations of positive and negative gaussian curvature

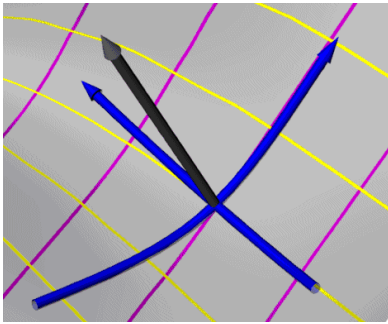


+

-

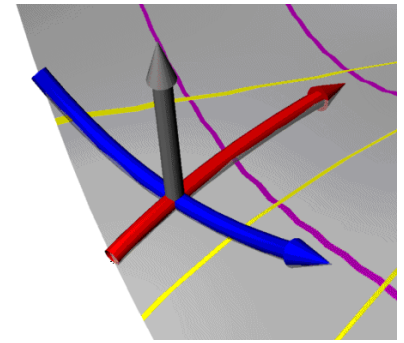


# Gaussian curvature: Definition and Meaning #2



+

-



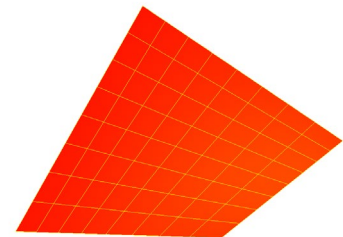
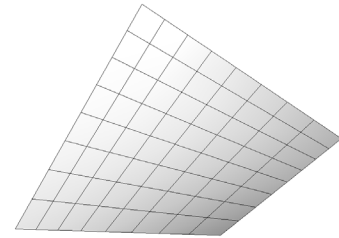
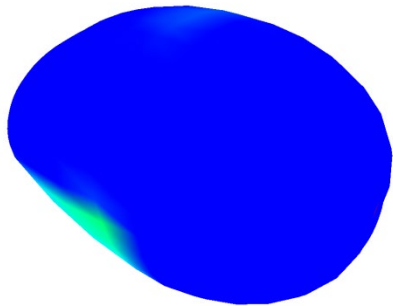
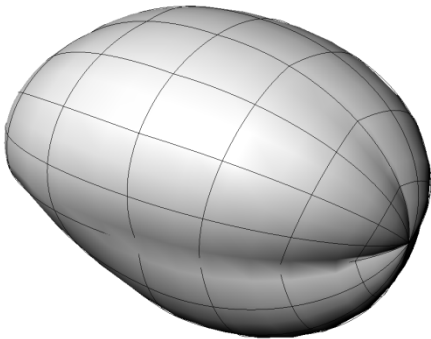
The “opening or closing of the surface”



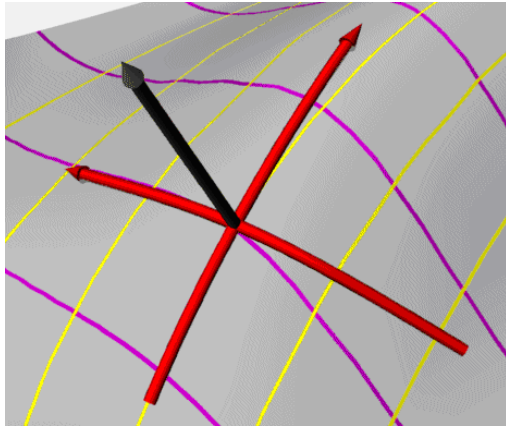
# Gaussian curvature: Definition and Meaning #2



The “opening or closing of the surface”

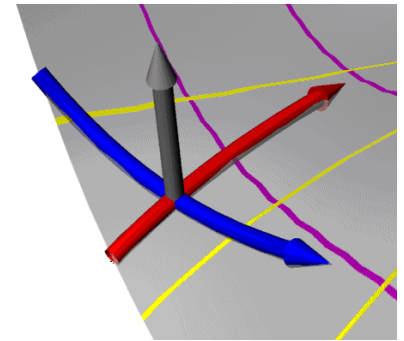


# Gaussian curvature: Definition and Meaning #2

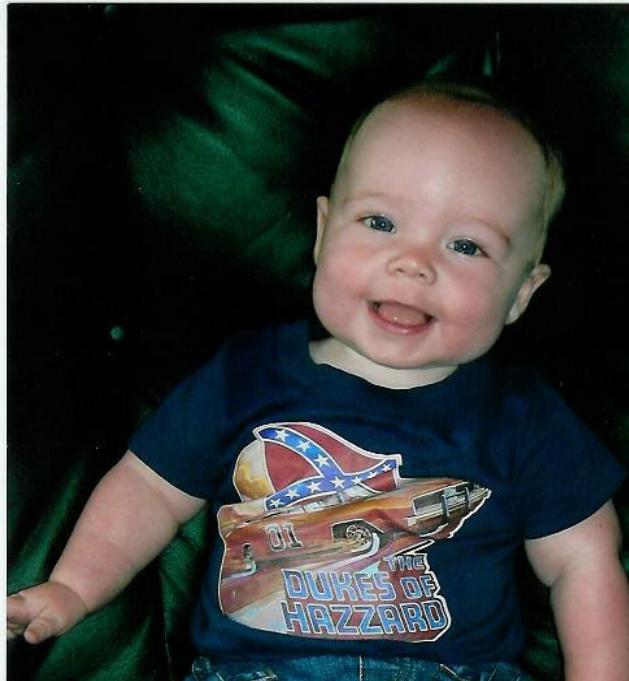


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-

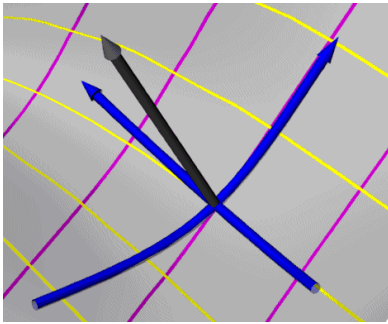


The “opening or closing of the surface”



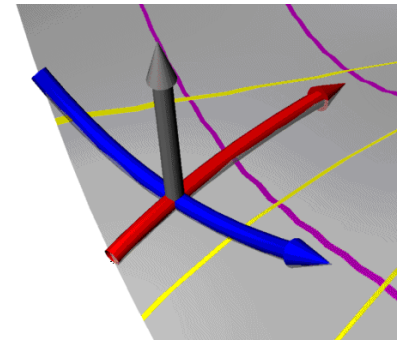


# Interpretations of positive and negative gaussian curvature



+

-

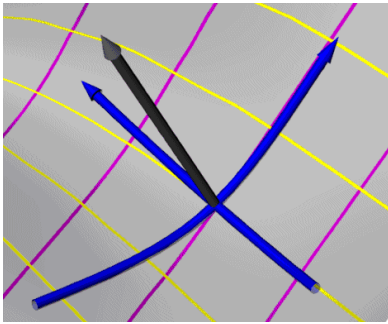


**Always positive**



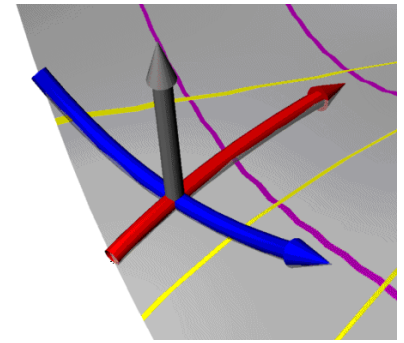
**Always negative**

# Gaussian curvature: Definition and Meaning #2

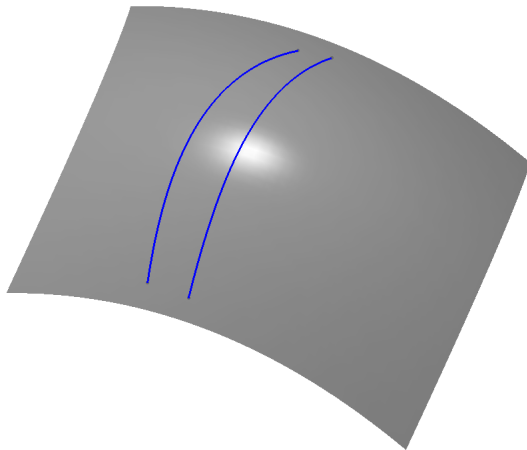


+

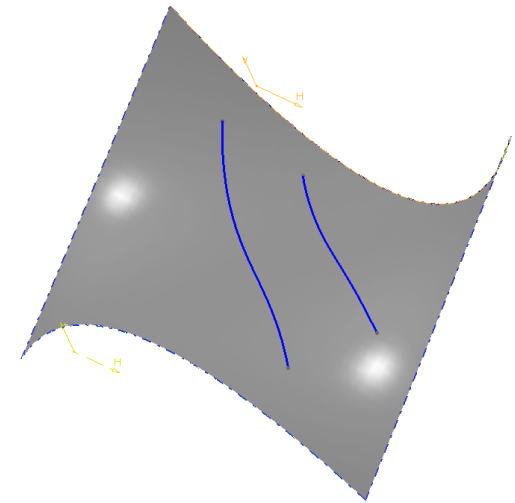
-



The “opening or closing of the surface”



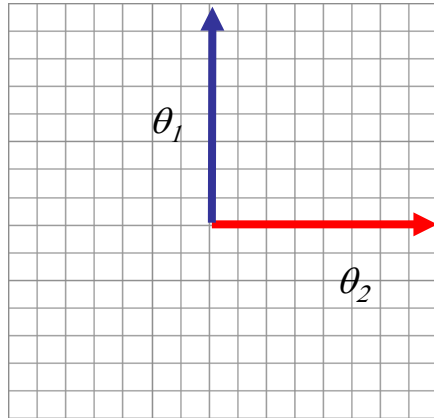
Parallel geodesics  
**converge**



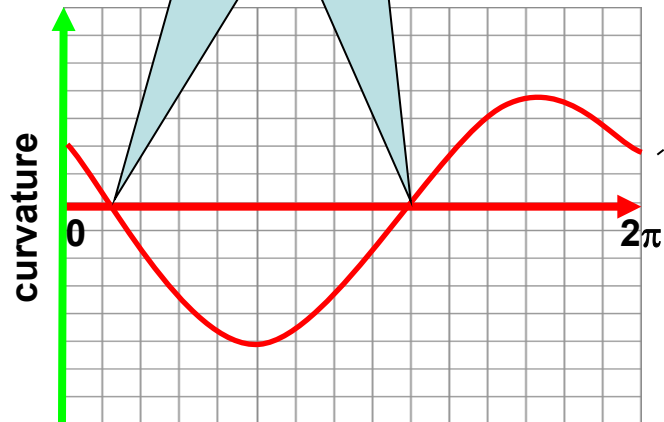
Parallel geodesics  
**diverge**



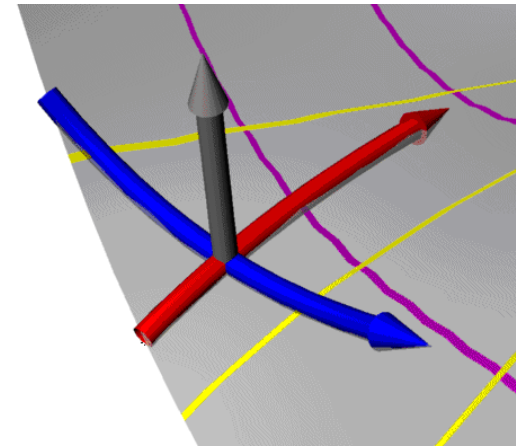
# Special note on negative curvature surfaces



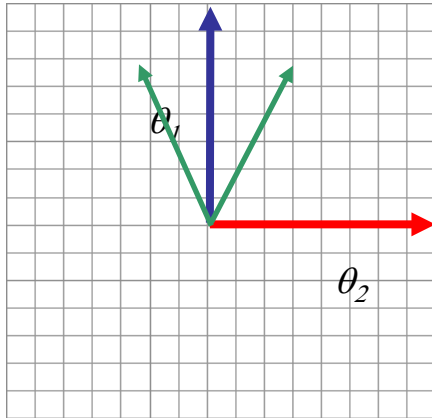
What's going on here??



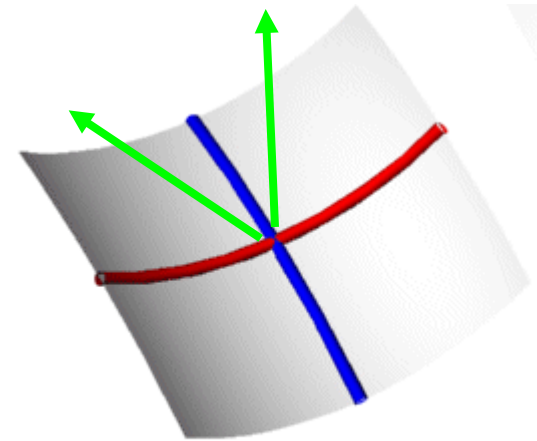
negative  
"saddle"



# Special note on negative curvature surfaces

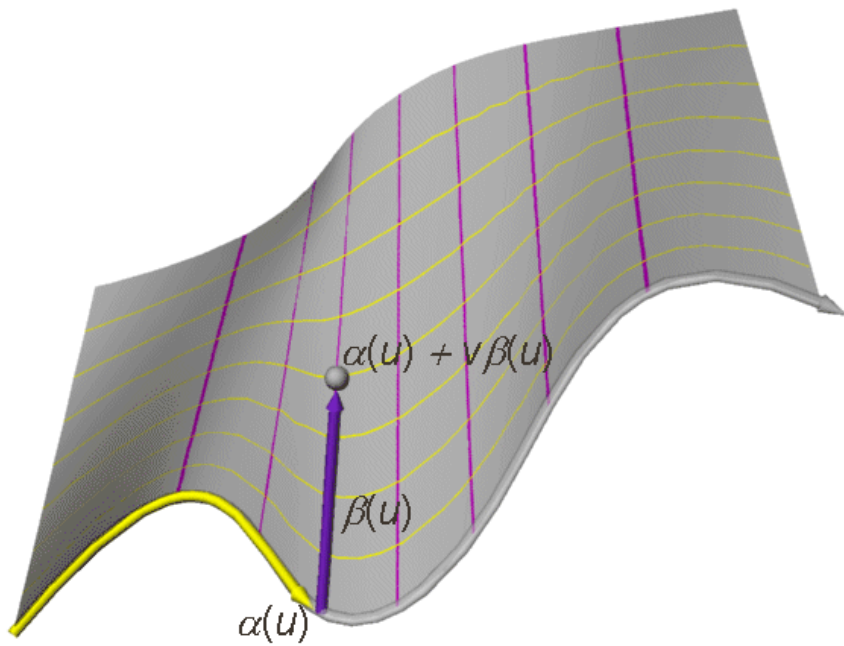


negative  
"saddle"

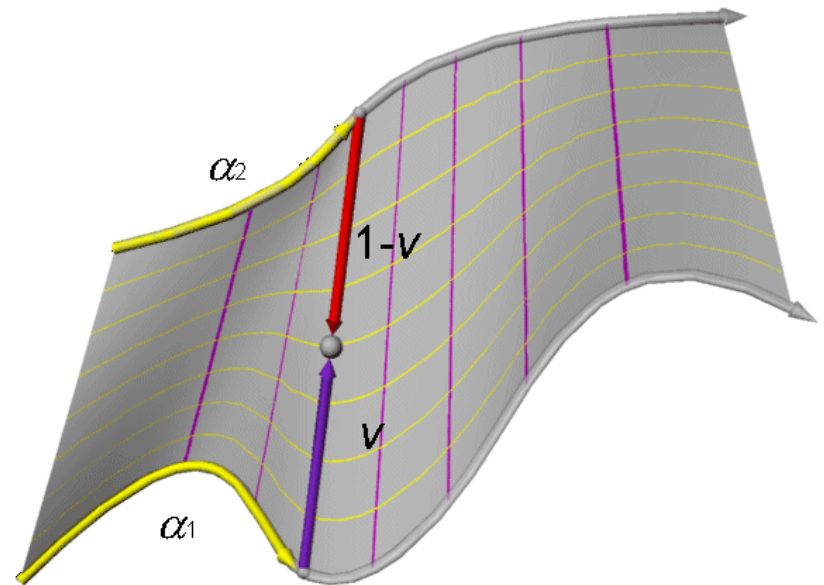


0 curvature,  
Asymptotic lines

# Ruled Surfaces

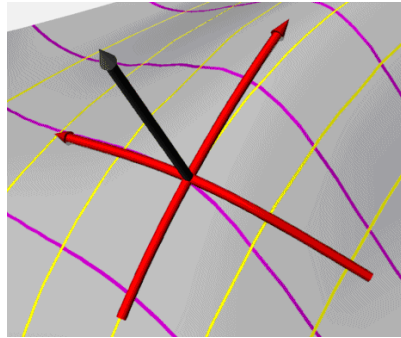
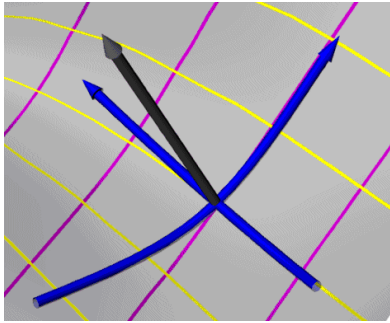


The envelope of a line along  
a curve in space

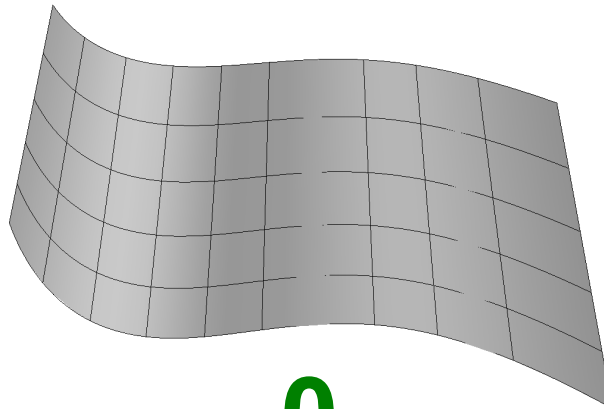


pointwise joining of  
2 curves

# Developable Surfaces

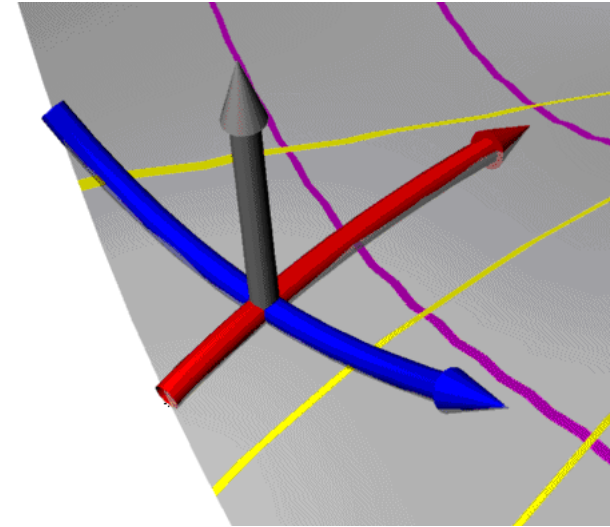


positive



0

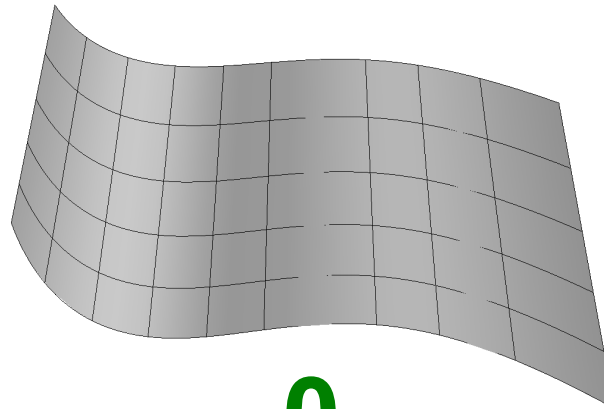
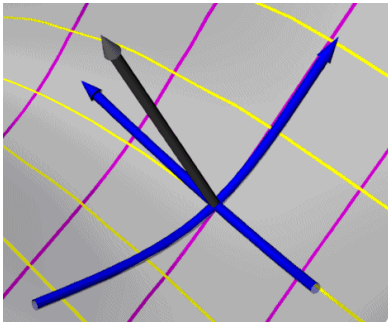
Always zero



negative

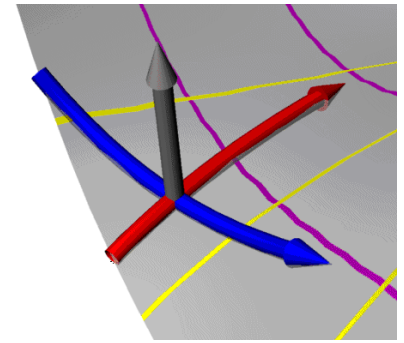
$$\begin{aligned} \kappa &\approx \left\| \begin{bmatrix} \hat{n} \cdot \sigma_{uu} & \hat{n} \cdot \sigma_{uv} \\ \hat{n} \cdot \sigma_{uv} & \hat{n} \cdot \sigma_{vv} \end{bmatrix} \right\| = 0 \\ &= (\hat{n} \cdot \sigma_{uu})(\hat{n} \cdot \sigma_{vv}) - (\hat{n} \cdot \sigma_{uv})^2 = 0 \end{aligned}$$

# Gaussian curvature: Definition and Meaning #2



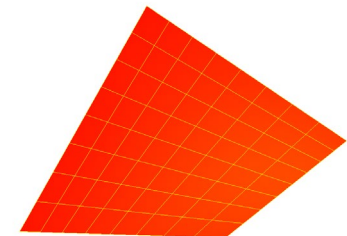
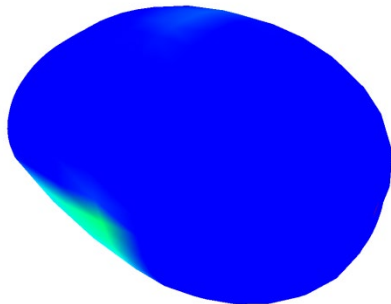
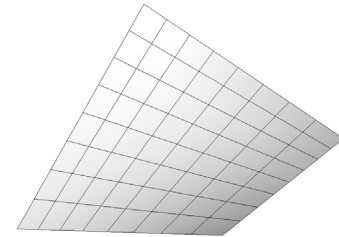
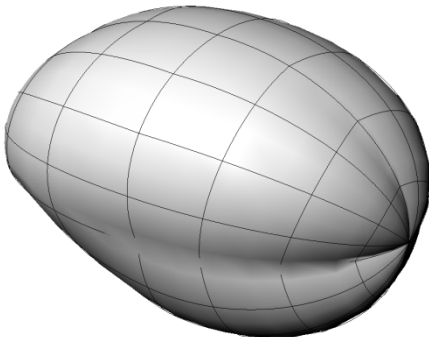
0

Always zero



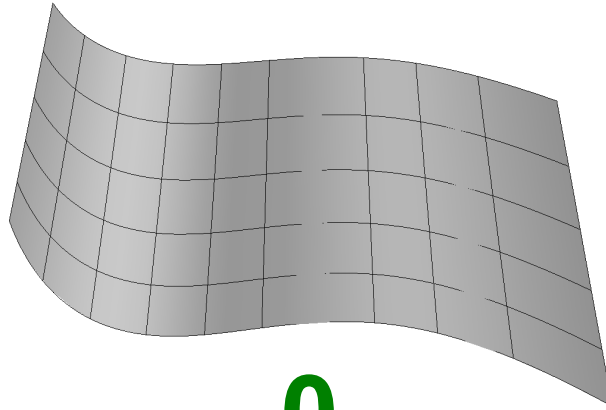
+

-



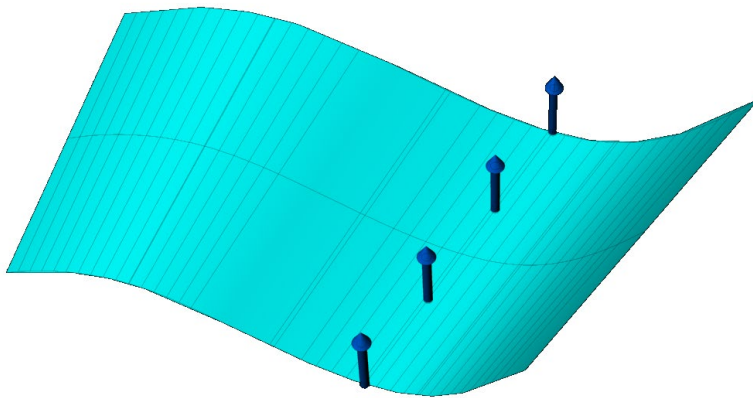
# Developable Surfaces

$$\begin{aligned} \kappa &\approx \left\| \begin{bmatrix} \hat{n} \cdot \sigma_{uu} & \hat{n} \cdot \sigma_{uv} \\ \hat{n} \cdot \sigma_{uv} & \hat{n} \cdot \sigma_{vv} \end{bmatrix} \right\| = 0 \\ &= (\hat{n} \cdot \sigma_{uu})(\hat{n} \cdot \sigma_{vv}) - (\hat{n} \cdot \sigma_{uv})^2 = 0 \end{aligned}$$

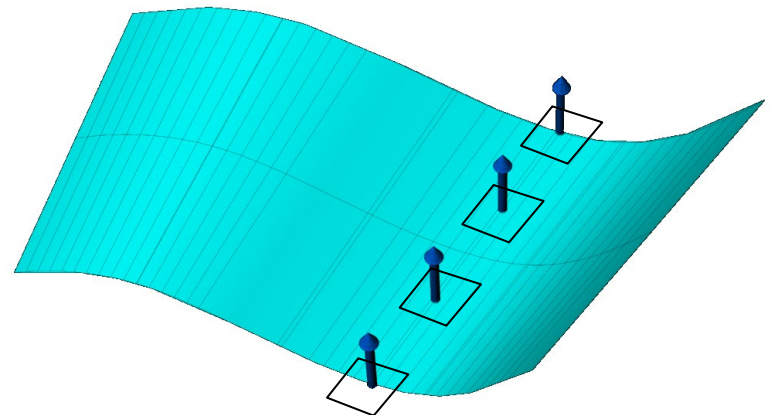


0

Always zero

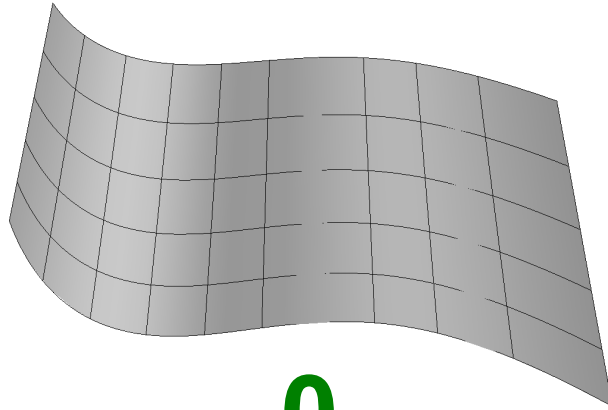


The surface normals  
are parallel



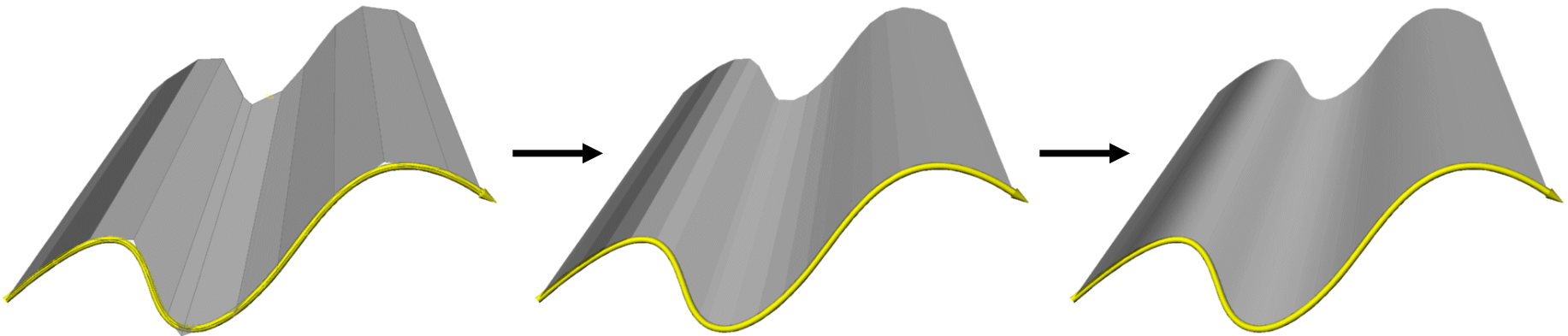
And lie in a  
common plane

# Developable Surfaces



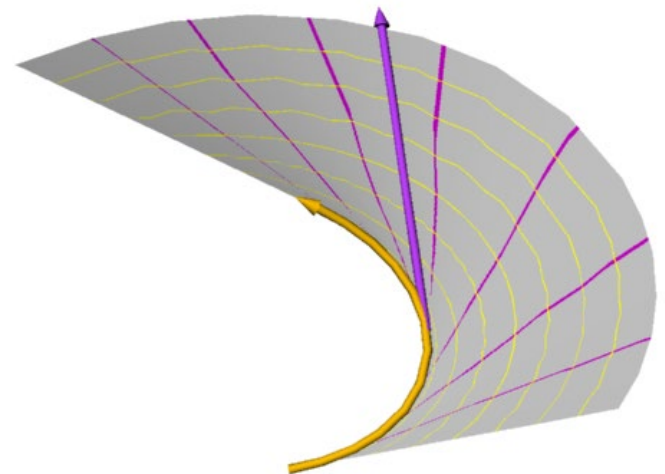
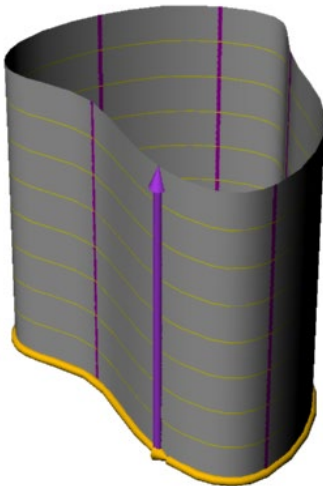
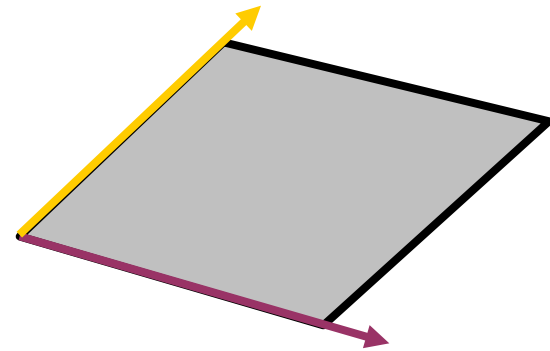
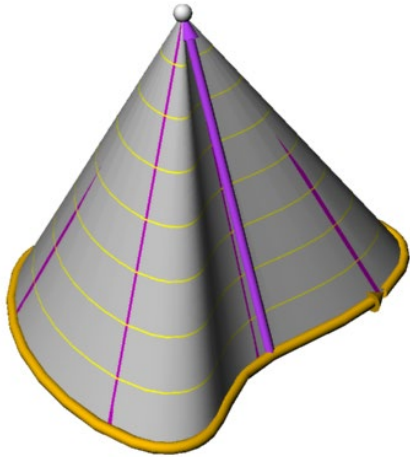
**0**

Always zero



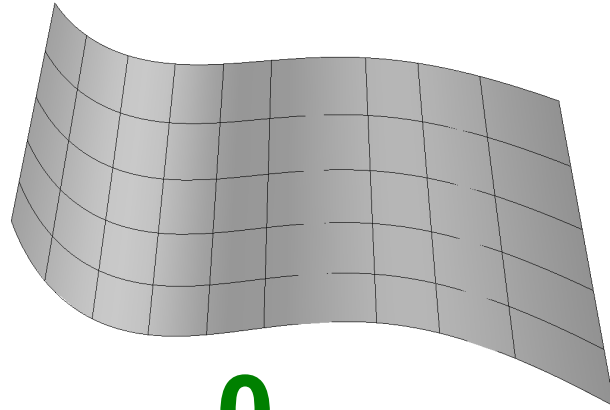
The limit of a family of planes

# Developable Surface Types



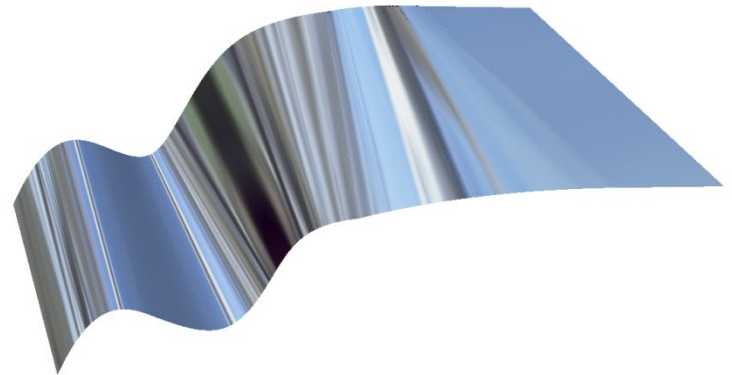
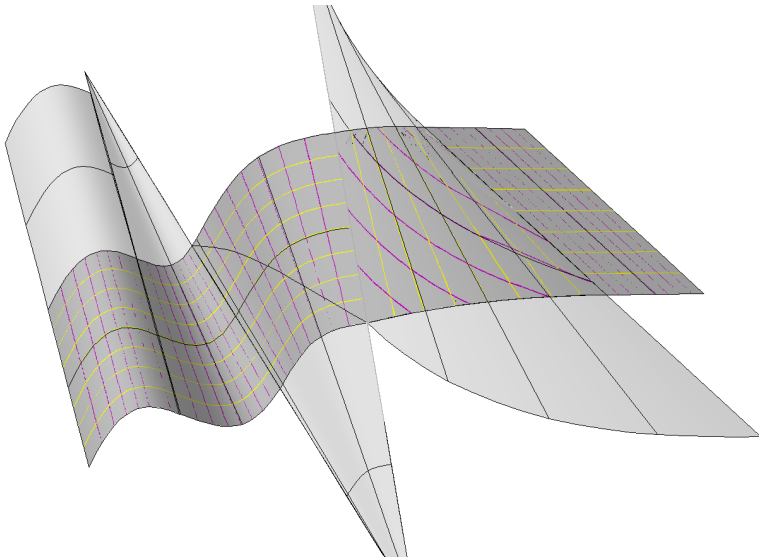


# Developable Surfaces

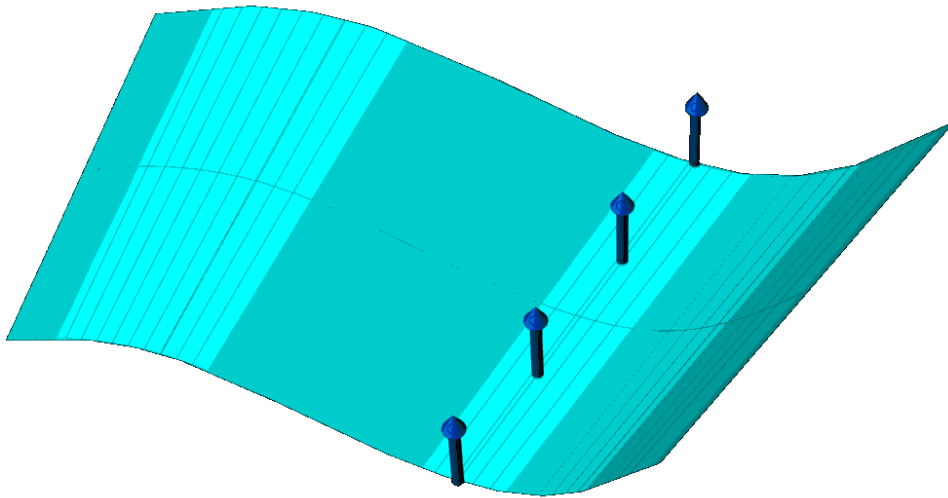


0

Always zero



# Ruled Surfaces



**The envelope of a line along  
a curve in space**

**pointwise joining of  
2 curves**

# Deformed Surfaces

